

Module 4: Financial Crises

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Lesson 4-0 Module 4 Information

[Lesson 4-0.1: Overview](#)



Hello and welcome, I hope you're doing well from wherever it is you're joining us. In this module, we focus on the role of central banks as lenders of last resort. For this purpose, we study several financial crises and discuss why they came about. We then consider the policy responses by central banks to this crisis. We see that central banks can succeed but that there are limits to how far they can go alone.

Financial crises

- Panic of 1907
- Great Depression
- Auto loans during 2008 crisis
- Causes?
 - Short-term or foreign debt
 - Bad long-term investments
 - Unsustainable government policies

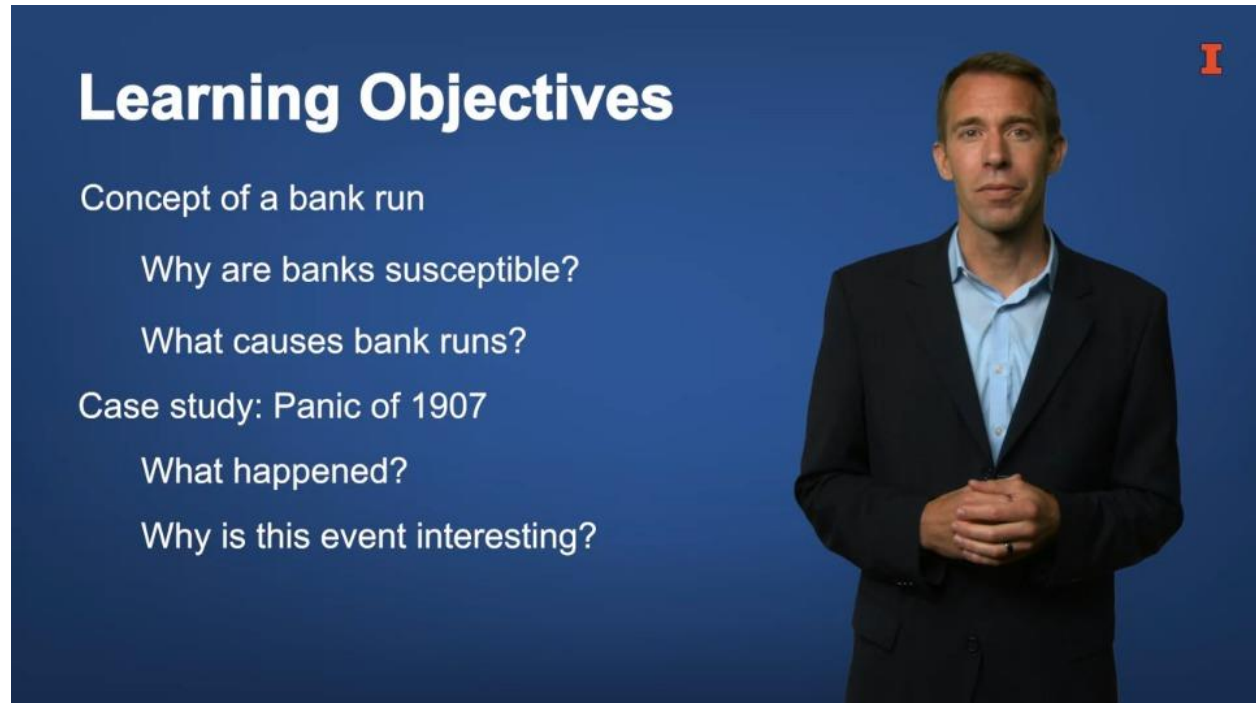
The first crisis we look at is the panic of 1907. This was the first Major financial crisis of the 20th century. It was also the last US. Banking panic that occurred before the Federal Reserve system was founded. This crisis helps us understand why a central bank as the lender of last resort is needed. We then study the Great Depression. The Great Depression is another useful example because much of the present-day social safety net did not exist. Next, we take an international perspective on financial crisis and identify common themes that help us identify the causes of financial crisis. This big picture perspective highlights credit booms that were financed with either short term or phone funding and combined with investments in long term projects or unsustainable government policies. Such credit booms foreshadowed financial crisis.

We reiterate the risks on relying on short term funding by studying the US auto loan market During the 2000 and eight financial crises here. Non-bank lenders relied heavily on short term funding markets to originate order loans when the short-term funding markets dried up. Auto sets collapsed. After all these examples, you may wonder whether credit booms always lead to financial crises and recessions. The short answer is no. As the next lesson discusses, it depends on in which sector the credit boom takes place. For example, credit booms in the manufacturing sector are often sustainable, at least historically. Having examined causes and consequences of financial crisis, we then turn to the policy responses. Specifically, how can central banks counteract bank runs and chart drops in credit supply. For this purpose, we studied the responses of the federal reserve to the 2008 financial crisis and the COVID-19 pandemic that illustrate the concept of the land of last resort, that is the Federal Reserve Lands when no one else bought. Monetary policy is not the only way to respond to financial crisis. Fiscal

policy, which involves government transfers and spending can also have the economy. In the final lesson, we therefore discussed the fiscal response to financial crisis.

Lesson 4-1: Understanding Financial Crises

Lesson 4-1.1. Bank Runs and the Panic of 1907



Learning Objectives

- Concept of a bank run
 - Why are banks susceptible?
 - What causes bank runs?
- Case study: Panic of 1907
 - What happened?
 - Why is this event interesting?

The slide features a blue background. On the right side, there is a photograph of Professor Ralf Meisenzahl, a man with short brown hair, wearing a dark blue suit jacket over a light blue button-down shirt. He is standing with his hands clasped in front of him. In the top right corner of the slide, there is a small red 'I' logo.

Hello and welcome to this class on bank runs and the Panic of 1907. We will proceed in two steps. First, we will understand what a bank run is and why banks are prone to runs. We will then consider our first major case study, the Panic of 1907. This is one of the most prominent financial crises in US history. It was a massive banking panic. It led to the formation of the modern Federal Reserve system. We will understand why the panic happened and how it was resolved. Understanding these historical episodes can be enjoyable and highly informative. Please don't worry about the precise historical details. I will mention many details, but these are mostly for fun, focus on the narrative and what we can learn.

Why Bank Runs?

Bank business model

- Borrow short (deposits)

- Lend long (loans)

Depositor runs

- First-come-first-served

- Last guy might get nothing



Let's get started, why are banks susceptible to runs? Well, banks borrow short term by taking deposits and lend long term this creates a maturity mismatch between assets, the loans, and the liabilities, the deposits. This maturity transformation is a key feature of the bank's business model. So why might a run occur? Remember that depositors can withdraw their funds at any time. If some shock or bad news arise this presents a risk to depositors. They might decide, play it safe and withdraw perhaps because they see people waiting in line in front of a bank. It is usually rational to join a bank run. If the saver joins the queue, then there is a chance that they get their money back. First in line will usually get paid. If they do not join the run, then they might lose out. Last in line might get nothing.

What Happens When Depositors Run?

Bank illiquid $\Rightarrow$ no more cash

Bank insolvent $\Rightarrow$ no more equity

Worst case: illiquid yet solvent bank fails



What are the consequences of bank runs? When many depositors arrive at the same time, the bank may not have enough cash on hand. Most deposits are tied up in long term investments. This is how maturity transformation works. In this case, the bank would be illiquid. It has run out of cash. A bank being illiquid is different from a bank being insolvent. An illiquid bank lacks cash, but its investments could still be profitable. An illiquid bank still has a viable business; it can still be solvent. On the other hand, an insolvent bank has run out of equity it is bankrupt. The worst-case scenario occurs when solvent banks fail due to runs. This can happen when banks are forced to sell assets with large discounts to meet depositor withdrawals. In our case studies, we will see how this dynamic plays out in practice. We will then see why various financial regulations and government institutions exist to prevent this worst-case scenario.

Case Study: Panic of 1907

Why interesting?

Two missing institutions:

1. No central bank
2. No deposit insurance



Let's now examine our case study the Panic of 1907. The Panic of 1907 is a helpful episode to learn about bank runs. At the time two features of the modern financial system were missing. First, there was no central bank. This means banks had no access to government-backed emergency loans. And second, there was no deposit insurance. Without deposit insurance, depositors were much more panicky. They could lose their life savings if their bank failed. This historical episode helps us to understand what might happen without the modern government interventions designed to prevent bank runs. Let's have a look how this financial crisis unfolded.

Panic of 1907: Causes

1. 1906 San Francisco earthquake
2. Poor harvest in 1907
3. Financial market speculation using bank loans



Like most crises, there was not a single cause, but a buildup of bad news that led to bank runs. The story starts with the San Francisco earthquake of 1906, which caused damages of about 1.5% of GDP. Such a large shock immediately rippled through the financial system. Insurers had to sell assets to meet insurance claims. Depositors withdrew money from banks, and stock markets in New York and London dropped. This trend continued. Over a series of days in March 1907 stocks declined by over 10%. Money markets were stressed, with interest rates shooting up from 2.5% to over 10% on several days. This event has been referred to as “the silent crash” because it did not happen in a single day, but over several days. The second shock was the 1907 crop harvest, it was not as rich as the previous year, reducing exports. The US treasury, unable to refinance maturing bonds, withdrew \$30 million from banks. With the government and depositors withdrawing more money from banks, the money supply contracted, credit became scarce, and interest rates further increased. In short, financial conditions worsened substantially during the summer of 1907. Against this backdrop, there was one more crucial factor. Speculators making risky bets in the stock market financed with short term bank loans. When these risky bets blew up, the bank loans defaulted, and this started the banking panic. This is the short version of the story. Here's the longer version which you may find fascinating.

Frederick Augustus Heinze



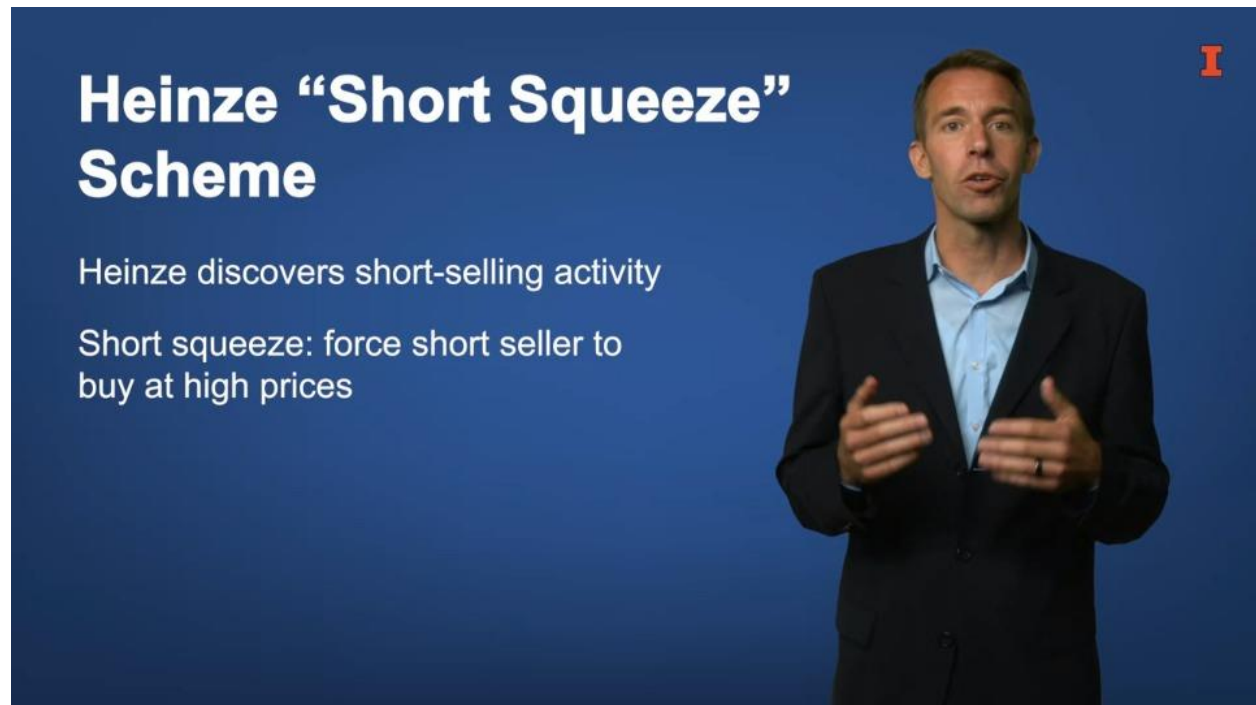
The Heinze process were very wealthy copper miners. In 1907, they moved to New York City and joined forces with Charles W. Morse to speculate in the copper equity market.

Charles W. Morse



They bought many shares of the United Copper Company. They used these stocks as collateral for their transaction with as many as 20 banking houses. The Heinze process

and their colleagues were speculators making risky bets on the stock market financed with short term bank loans.

A video frame with a blue background. On the right, a man in a dark suit and light blue shirt is speaking, gesturing with his hands. On the left, white text reads: "Heinze 'Short Squeeze' Scheme", "Heinze discovers short-selling activity", and "Short squeeze: force short seller to buy at high prices". A small red "I" logo is in the top right corner.

Heinze "Short Squeeze" Scheme

Heinze discovers short-selling activity

Short squeeze: force short seller to buy at high prices

On October 9th, 1907, Otto Heinze decided to analyze the trading behavior in the United Copper stock. He discovered that about 450,000 shares of this stock were traded. Even so, according to the company, only 425,000 shares were outstanding. That meant that some brokers had lent their stocks to investors who shorted the stock. This means these other investors borrowed the stock and sold it in the hope to buy it back at a cheaper price later and thereby make a profit. The Heinze suppressor believed that the brokers had lent out the shares that they owned and had pledged as collateral. They reasoned that by calling in the shares they could create a situation in which the short sellers would be forced to buy the shares they owned at a very high price. This is called a short squeeze.



To induce the short squeeze, the Heinze process had to borrow \$1.5 million to pay off the loans collateralized with the stocks. When the United Copper stock dropped again from \$45.50 to \$37.75, the process started calling in the shares by requesting to pay off their loans. Within hours, the stock rose by \$23, but then declined to about \$52. However, the process had miscalculated. All 20 banks that had extended the process loans backed by the stocks delivered all the stocks. As a result, the process was not able to meet all the required loan payments and refused to accept the delivery of the stock. The lenders immediately sold off the rejected collateral and the stock price tumbled. After three days, the United Copper stock had lost 50% of its value and eventually traded at \$10 per share. The short squeeze scheme had failed miserably. The Heinze brothers were bankrupt. On October 16th, their brokerage firm closed. So did a small bank owned by one of the Heinze brothers. That small bank, however, used the New York based Mercantile Bank for payment services and had deposits worth about \$1 million. The Mercantile Bank also had direct exposure to the United Copper stock, that had collapsed in value. These events were so alarming that people began pulling their money from any bank associated with the Heinze process. The panic began.



To coordinate an orderly settlement of all claims, the New York Clearing House, a consortium of New York banks, stepped in on October 20th. This was a private sector intervention designed to calm the banking panic.

Private Sector Intervention

New York Clearing House

- Largest banks are members
- Settle disputes

Problem: Knickerbocker Trust Company



However, during the clearinghouse meetings, the Knickerbocker Trust, which was the third largest bank in New York, was implicated in financial irregularities due to their association with the Heinze brothers. This signaled that the Knickerbocker Trust was in

serious trouble. It is not clear that the Knickerbocker Trust was financially unsound, but banks rely on trust. If depositors question whether they will get their money back, they simply will withdraw.



And so, on the morning of October 22nd, 1907, a line started forming in front of the Knickerbocker Trust office an hour before they opened. The line seems to grow by the minute, with all depositors demanding their money back. The police were called to help with the line formation. By midday, Knickerbocker announced that it had assets worth about 69 million and liabilities of only 64 million and could pay out everyone. No one dropped out of line. Knickerbocker lost 8 million of deposits that day. This is an example of a classic bank run. Knickerbocker was financially sound, but worried depositors demanded their money back anyway.

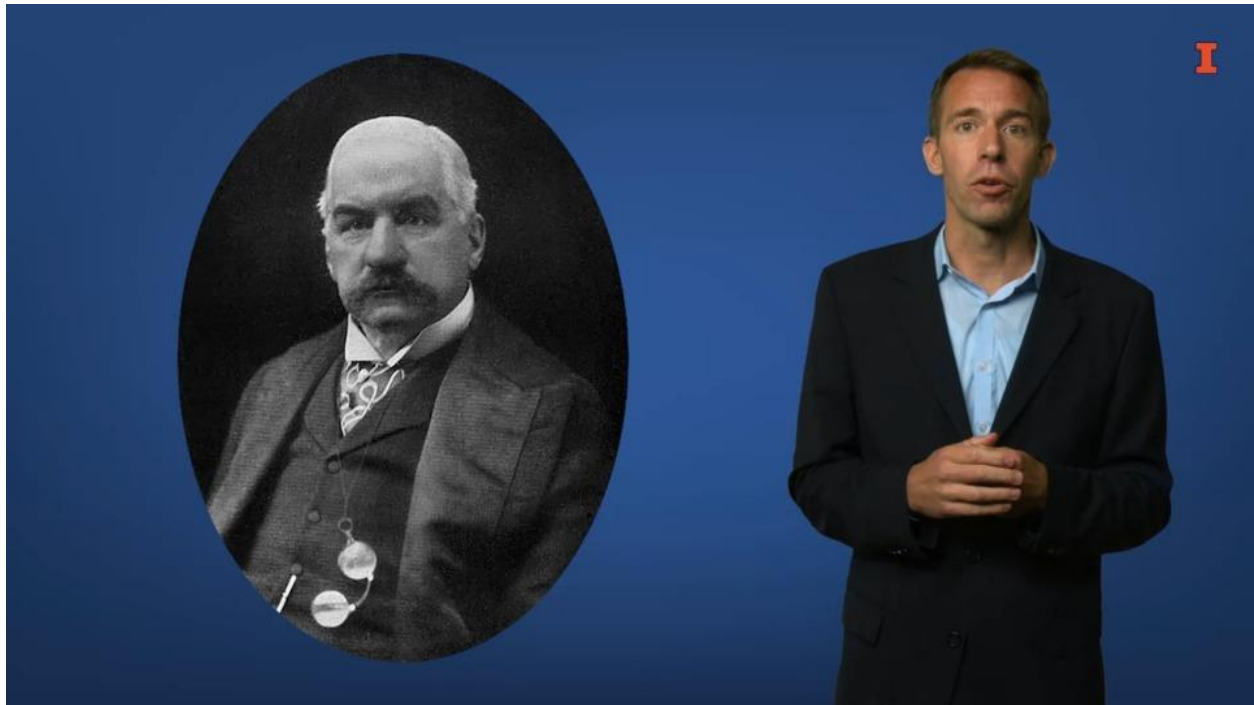
Private Sector Intervention

Panic spreads to other banks and NYSE

Private banks offer cash loans
(including J.P. Morgan)



The next day, Knickerbocker ceased operations. However, many other healthy banks in New York were swept up in the panic. By October 24th, 1907, several banks had failed. When runs on the Trust Company of America started, New York's most famous banker, J.P. Morgan, and others stepped in. They provided cash loans to meet depositor demands. The lack of cash pushed the New York Stock Exchange to the brink of collapse. Once again, J.P. Morgan intervened, raising the capital needed to keep the stock exchange in business.



With more banks on the brink of collapse, J.P. Morgan summoned about 120 New York bank presidents to his home in New York in early November. He demanded that they help solve the crisis. To force a solution, he famously locked the banker into his library. Eventually, the bankers agreed, and J.P. Morgan contributed another \$25 million to resolve the crisis. This good news stopped the banking panic, but plenty of the damage was already done. It was too late to prevent bank failures and too late to stop a severe recession.

Summary

Case study: The Panic of 1907

Worried depositors ran on banks

⇒ Need deposit insurance

Cash loans can stop bank runs

⇒ Need government-backed lender
of last resort



Let's wrap up, The Panic of 1907 is viewed as one of the most severe financial crises in US history. What can we learn from this episode? First, depositors panicked because they were worried about losing their savings. A lack of trust and cash can lead to depositors' panic. This explains why deposit insurance prevents bank runs; insured depositors will be protected from any potential losses. This is probably why classical bank runs with depositors lining up around the block are rare nowadays. The second lesson is that cash loans can help stop bank runs that are in progress by meeting depositor demands. In the panic of 1907, J.P. Morgan acted as a lender of last resort. In 1907, we were fortunate that J.P. Morgan did not run out of cash or face any trouble at his own bank. Without government backing, the option of private lenders of last resorts are limited. This explains why the Federal Reserve, with its unlimited resources, can be more effective as a lender of last resort.

Lesson 4-1: Understanding Financial Crises

[Lesson 4-1.2. The Great Depression](#)

Learning Objectives

Case study: Great Depression

Financial and economic collapse

Federal Reserve ineffective—why?

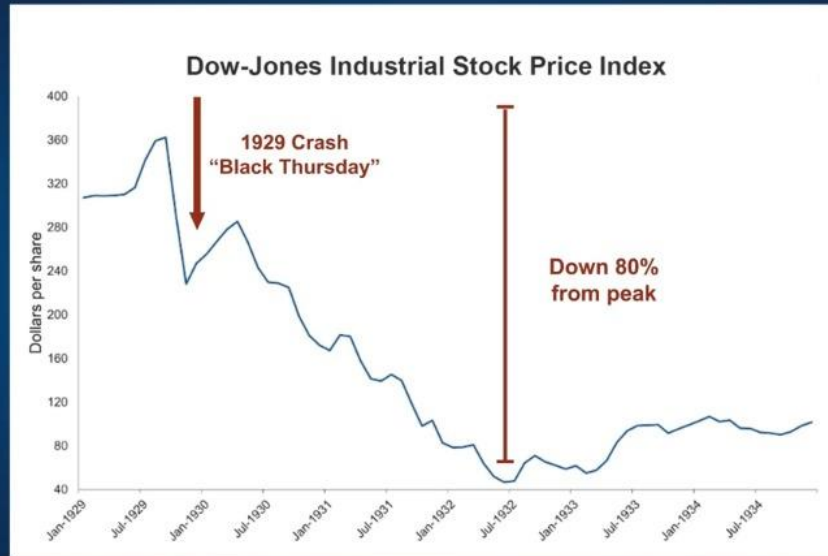
Which policies worked?

Hello and welcome to this class on the Great Depression. In the modern era, the Great Depression was the most severe financial crisis and economic downturn in the United States and most of the western world. We will examine how financial conditions evolved during the Great Depression. We will see the massive number of bank failures. We will understand why the Federal Reserve system was ineffective at stopping the banking panic. Last, we will examine the large-scale government response that kick started the economy and restored confidence in the banking sector. This included government spending programs. It also included the creation of new institutions such as deposit insurance. Understanding these historical episodes can be enjoyable and highly informative. Please do not worry about the precise historical details. I will mention many details, but these are mostly for fun. Focus on the narrative and what we can learn. The Great Depression is interesting because most policies that would mitigate the effects of a financial crisis, such as deposit insurance or unemployment benefits, did not exist in the US. This helps us to understand what would happen in an economy if the government did not intervene during a financial crisis.

Stock Market Crash in 1929

- October 24, 1929 ⇒ “Black Thursday”
- Financial market speculation on credit unwinds
- Initial losses amplified

Let's start by briefly summarizing the start of the Great Depression. On October 24, 1929, the so-called Black Thursday, the market lost 11% on the opening bell. The crash followed a drop of the London Stock Exchange caused by fraud allegations against large investors. On October 28, 1929, more and more investors faced margin calls on stocks bought on credit due to dropping stock prices. As a result, more investors sold their shares and cut their losses. The Dow lost almost 13% that day. The next day, investors traded a record number of shares. 16 million shares changed hands on a single day. The next day, the stock market showed signs of illiquidity. Some stocks could not be sold at any price. The Dow lost another 12%.



The chart shows the Dow-Jones industrial stock price index. As you can see, the developments over October 1929 were just the prelude to long decline in stock prices. From September 1929 to July 1932, stock prices chopped by over 80%.

Collapse of Banking System

Bank risks

- Direct exposure to stock market crash
- 16,000 banks had no access to lender of last resort

Bank failures

- Contagion from small to large banks
- Local bank runs (not national)
- 9,000 failures

When the crisis began, over 8,000 commercial banks belonged to the Federal Reserve system, but nearly 16,000 did not. At the time, banks often owned stocks themselves and had limited cash reserves, so not having access to the Federal Reserve system for

cash was problematic. Banks without access to the Federal Reserve often deposited cash with other larger banks, creating the potential for contagion in case small banks withdrew their deposits from large banks. Or large banks themselves ran into difficulties, in this environment, depositors naturally wondered about the soundness of banks. A perhaps surprising feature of the Great Depression is that there was not one national bank panic, but a large number of local banks runs. In total, about 9,000 banks closed their doors between 1929 and 1933.

Money Supply Contracts

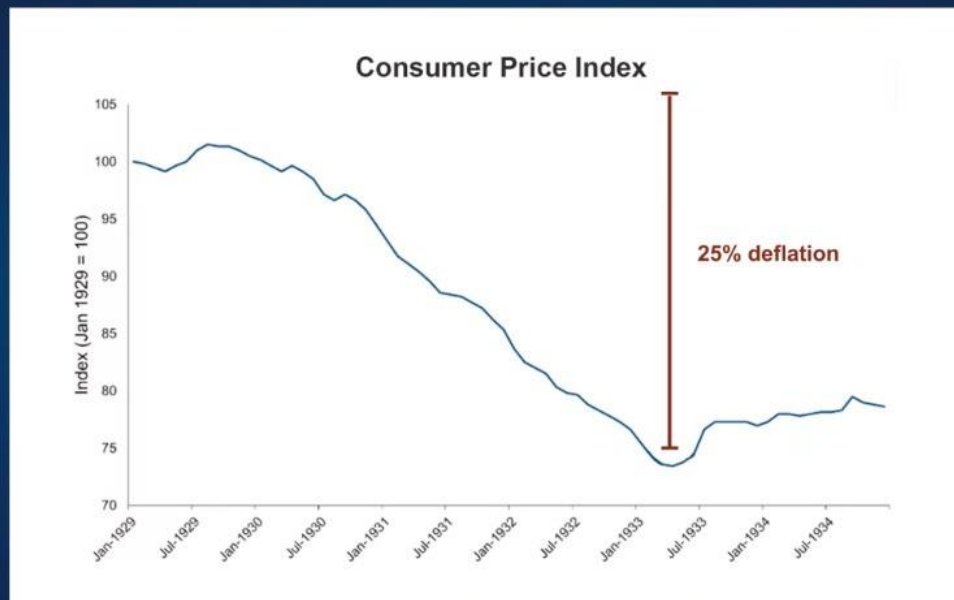
Collapse of bank deposits

Banks cut lending

Gold standard limited Federal Reserve

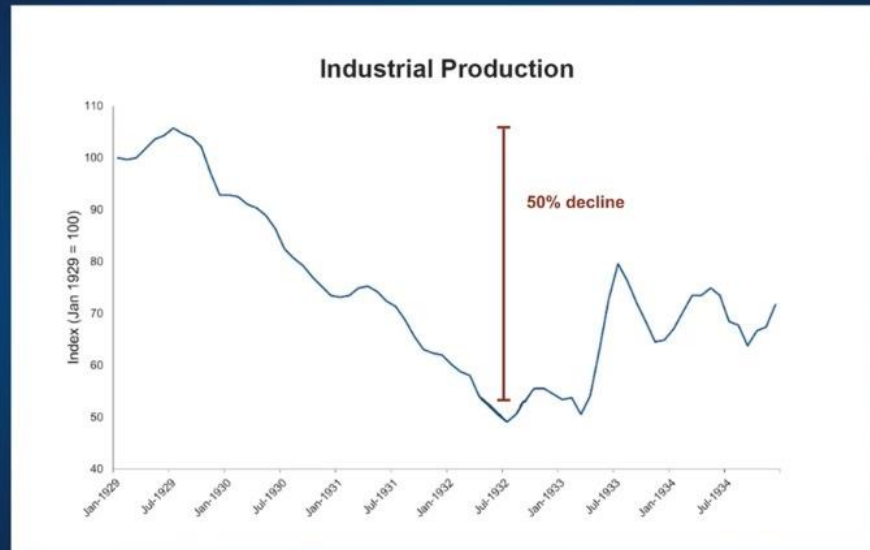
Could not increase money

The large number of bank closures sharply reduced deposits. Banks in turn tried to shore up their cash reserves and reduced lending. And as a result, the money supply shrank. You may wonder why the Federal Reserve did not just increase the money supply. At this time, the United States, as most other nations, was on the gold standard, meaning that money was packed to gold at a fixed exchange rate that severely limited the central bank's options.



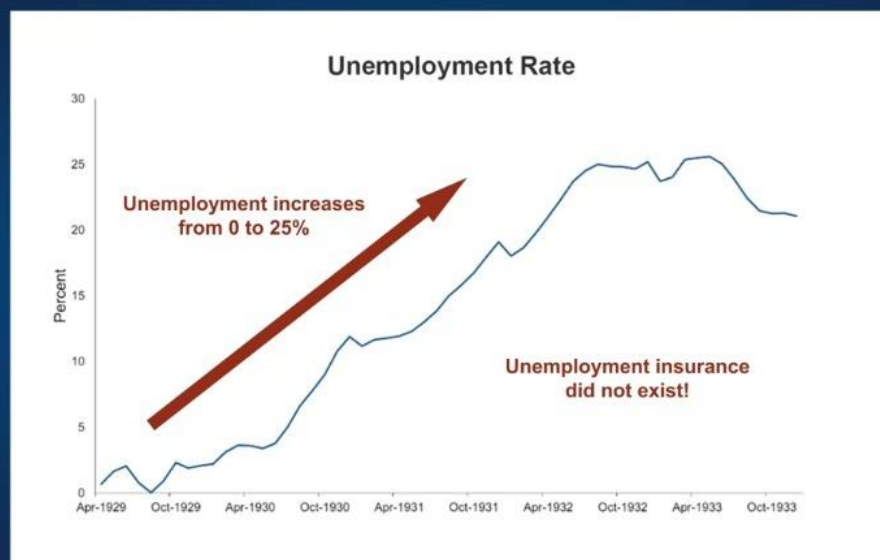
National Bureau of Economic Research, n.d.

Lower money supply put downwards pressure on prices. In this chart we can see that consumer prices deflated by about 25% during the Great Depression. The decline was even more pronounced for food. This deflation meant that farmers and firms faced higher debt in real terms, meaning in terms of goods produced. As a result, many farmers and businesses went into bankruptcy. This bankruptcy would cause additional stress in the banking system. Lower prices and especially expected lower prices in the future reduce the incentives to invest. If holding cash means that you can buy more tomorrow in terms of goods, why invest the cash in a risky business endeavor that can only charge low prices in the future?



Board of Governors of the Federal Reserve System, n.d.

With deflation setting in and less credit being available, business investment plummeted. The reduction in investment, together with the large number of bankruptcies, led to a prolonged decline in industrial production. As you can see in this chart, industrial output dropped by 50% during the Great Depression. This was an unprecedented drop-in economic activity.



National Bureau of Economic Research, n.d.

With the economy contracting, unemployment rose fast. In this chart you see the

unemployment rate increased from virtually zero at the beginning of the depression to over 25% by the end of the depression. Poverty and social unrest followed as unemployment insurance did not exist.

“Bonus Expeditionary Forces” in 1932

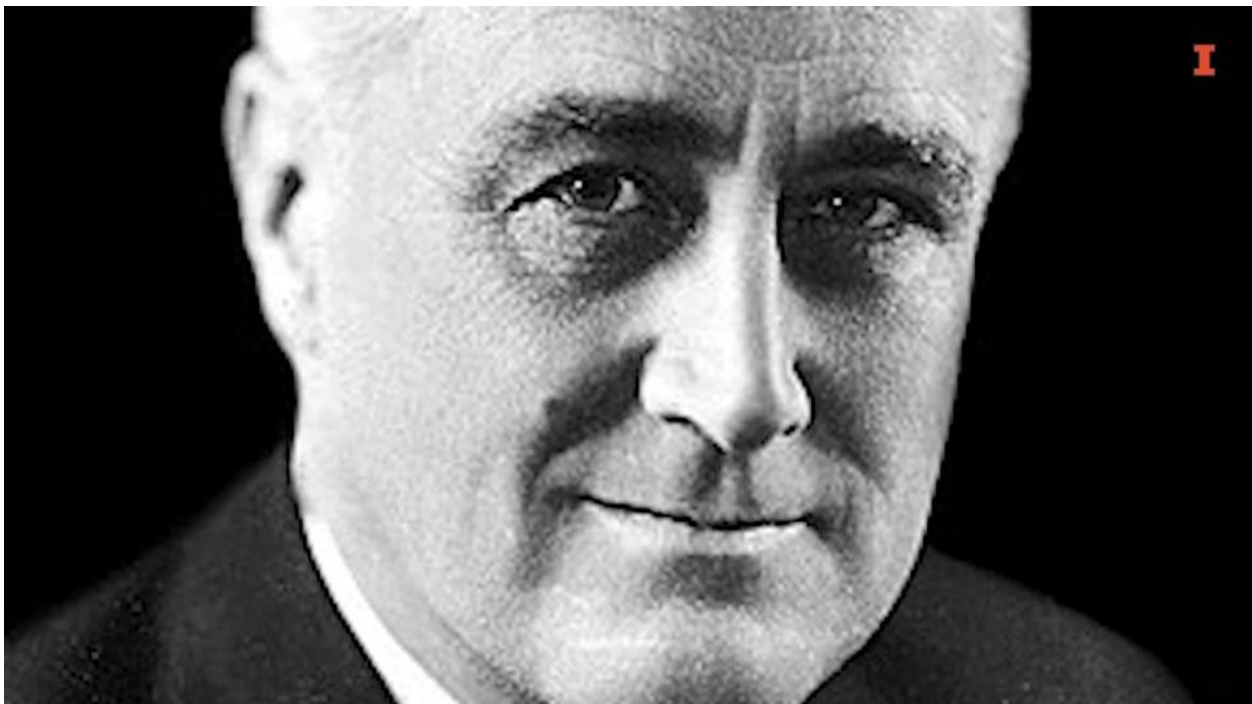


One of the most prominent examples of social unrest was the World War I Veterans March. In 1932, the [inaudible] World War I veterans organized a group called the Bonus Expeditionary Forces to march on Washington DC.

“Bonus Expeditionary Forces” in 1932



An estimated 20,000 veterans converged on the nation's capital and camped in Washington DC, demanding a bonus payment that had been promised but had stalled in conquest. The protest was eventually broken up by force, but the pictures of tanks dispersing hungry veterans weakened President Hoover's reelection bid.



In November, Franklin Delano Roosevelt's campaign on economic populism won the election, and the veteran bonus was eventually paid in 1936.

Government Intervention

Decisive actions by President Roosevelt

Emergency Banking Act of 1933

Reopen only solvent banks

Federal Reserve “guaranteed” deposits

One of the first actions of Roosevelt after being sworn in was to declare a bank holiday, or more precisely, a bank holiday week on March 6, 1933. For an entire week, nobody had access to banks or banking services. That meant that no money could be withdrawn or transferred, and arrested bank runs that had started again earlier that year. The purpose of the bank holiday was to buy time to enact legislation that would restore confidence in the banking system and thereby prevent future runs. The Emergency Banking Act of 1933 achieved this goal. In short, the act would ensure that only banks in good condition would reopen and banks in poor conditions would be shut down. Moreover, the act gave the Federal Reserve more authority and made it easier to lend to banks. The Federal Reserve effectively became the guarantor of deposits of reopened banks.



The results were immediately visible. After the act passed, the Dow-Jones Industrial Stock Price index gained over 15% in one day, and bank runs ceased.

Government Intervention

Decisive actions by President Roosevelt

Glass-Steagall Act of 1933

Separated deposits from risky investment banking

Established deposit insurance

The Banking act of 1933, also called the Glass-Steagall Act, added protections for depositors in June 1933. By separating commercial meaning, deposit taking, and investment banking, depositors were no longer exposed to banks risk taking in stock or more generally, securities markets. The act also established the Federal Deposit

Insurance Corporation, or FDIC, providing some insurance for deposits, which helped to prevent further runs. The Banking act also increased bank supervision and altered some features of the Federal Reserve system. While the banking panic was over, the real economy was still in dire straits.

Government Intervention

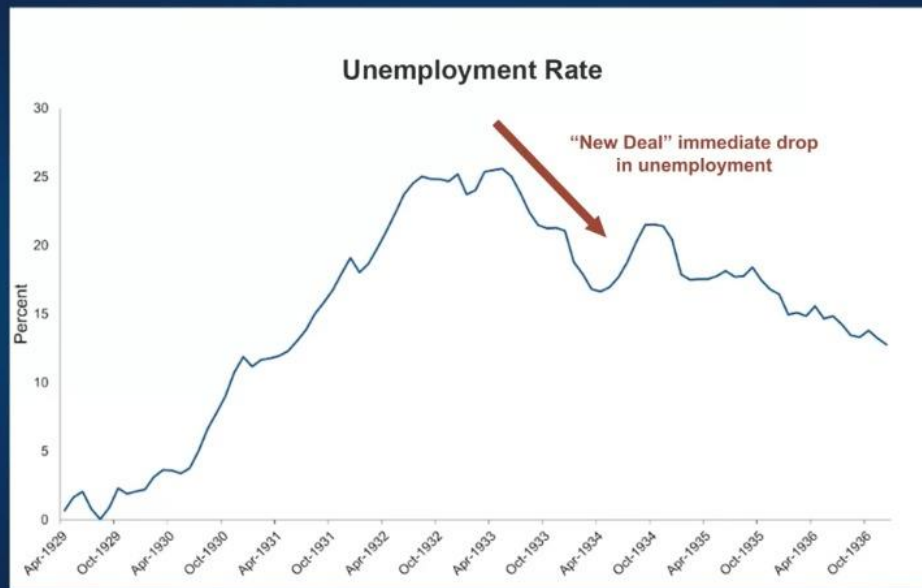
Decisive actions by President Roosevelt

“New Deal” for workers

Works Progress Administration

Civilian Conservation Corps

As part of the “New Deal” that fundamentally reshaped the role of government in economic crisis, Roosevelt's first objective was to alleviate the suffering of unemployed workers. The government created jobs through the Works Progress Administration and the Civilian Conservation Corps. The Works Progress Administration gave some 8.5 million people jobs, mostly in construction. More than 650,000 miles of roads, 125,000 public buildings, 75,000 bridges, and 8,000 parks were built. The Civilian Conservation Corps provided work that included planting trees, building flood barriers, fighting forest fires, and maintaining forest roads. It is important to note that these projects could be started on short notice and had immediate effects.



National Bureau of Economic Research, n.d.

The chart shows you that unemployment dropped by 9 percentage points within a year.

Government Intervention

Decisive actions by President Roosevelt

“New Deal” for workers

Works Progress Administration

Civilian Conservation Corps

Government “safety net”

National Labor Relations Board

The economic plight of people had also shown the need for a social safety net. The creation of a pension system through the Social Security Administration. Unemployment benefits and disability benefits were part of the “New Deal”. The “New Deal” also restructured labor relations, by creating the National Labor Relations Board. The economy seemed to be on track again. However, the “New Deal” needed to be

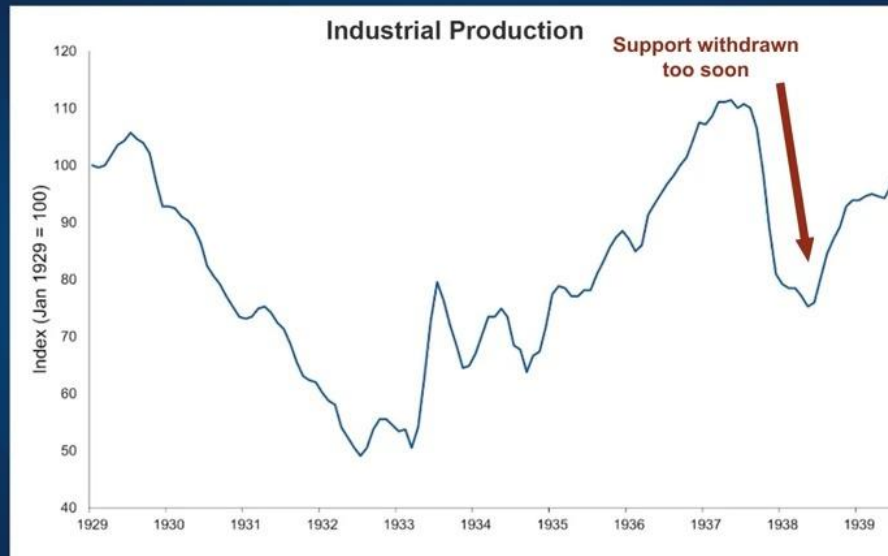
financed, and taxes increased. Moreover, with the improvements in the economy, the fiscal stimulus was withdrawn. At the same time, the Federal Reserve increased reserve requirements for banks, thereby reducing the availability of credit.

Policy Mistakes in 1937

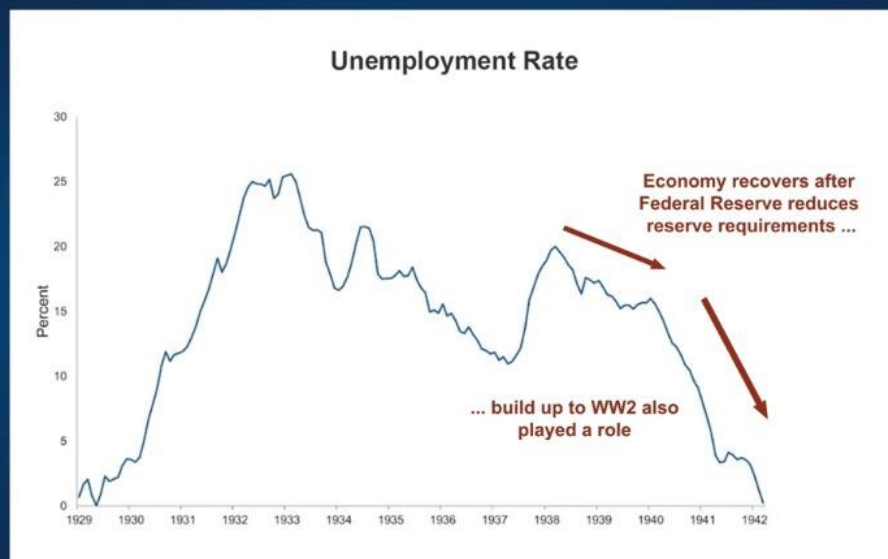
Higher taxes to fund New Deal

Federal Reserve increased reserve requirements

In other words, contractionary monetary and fiscal policy hit the economy at the same time in 1937.



You can see the result in the chart. Industrial production collapsed again in 1937. The recession ended when the Federal Reserve reduced reserve requirements. The 1937 episode provides a cautionary tale against withdrawing economic support too early.



The recovery was swift as you can see in this chart, unemployment chopped fast after the 1937 recession. However, it is important to note that another stimulus helped the economy to grow so fast. The defense buildup in Europe and the subsequent increase

in government spending. You can see that starting mid 1940, the reduction in unemployment accelerated.



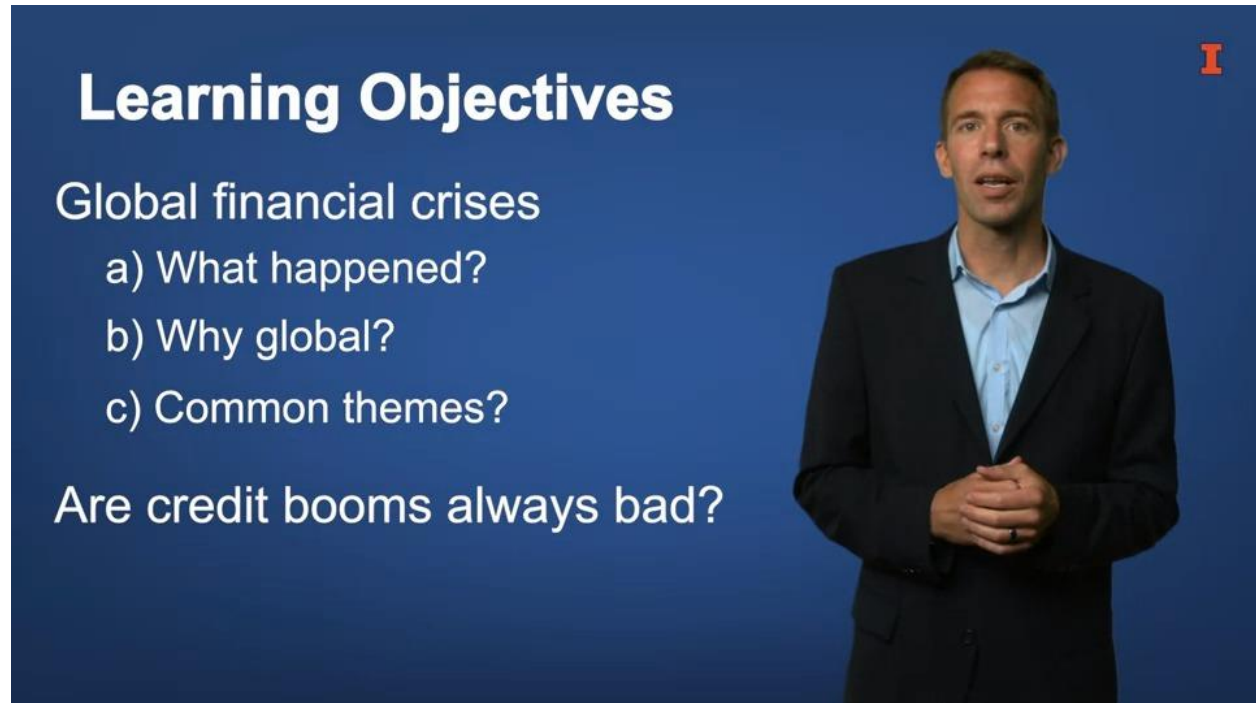
Summary

- Great Depression as worst-case scenario
- Lack of government safety net
- Only large-scale government interventions can restore confidence
- Early withdrawal of support counterproductive

What have we learned in this lesson? First, using the Great Depression as a case study of the consequences of financial crisis. We saw that the effects can be devastating, especially in the absence of a social safety net or government interventions. Second, only large-scale changes to the financial system can restore the confidence in the financial system. Third, the recession of 1977 illustrates that providing too little or withdrawing government support for the economy too early may result in another economic slowdown.

Lesson 4-1: Understanding Financial Crises

Lesson 4-1.3. Common Features of Financial Crises



Learning Objectives

Global financial crises

- a) What happened?
- b) Why global?
- c) Common themes?

Are credit booms always bad?

Hello, and welcome to this class on common features of financial crises. We will provide a broad overview of the major global financial crises over the past 150 years. Our goal is to understand how the crises unfolded, why the crisis was global, and are there any similarities across these crises? We will uncover these common themes of financial crisis. You will see that excessive debt always plays a role. There's always a run on short-term debt such as bank deposits. Unsustainable long-term investments or government policies often play a role as well. Does a rapid expansion of credit always end in a financial crisis? Towards the end of the class, we will address this question as well. Let's get started.

Major Global Financial Crises

1. Panic of 1873
2. Barings Crisis of 1890
3. Panic of 1907
4. Great Depression of 1930s
5. Financial Crisis of 2008



The five most serious global financial crisis over the last 150 years were the Panic of 1873, the Barings Crisis of 1890, the Panic of 1907, the Great Depression of the 1930s, and the 2008 Financial Crisis. Let's consider each of them in turn. This will be a brief description and remember, our goal is to find the features that come up time and time again. Please try to focus on the big picture and not on all the small details.

Panic of 1873



The Panic of 1873 was the first global banking crisis. Railroads were the hot investment

of the mid 1800s. Milestones like the completion of the Transcontinental railroad connecting the US east and west coast in 1869, fueled the interest in railway investment.

Panic of 1873

Rapid growth of railroad investing
around the world



This was not only true in the US, but also in Europe. For instance, railway companies financed by debt popped up all over Germany. Railroad stock prices jumped worldwide. Many banks held railway bonds as an investment. A key vulnerability of a country's banking system became railroad bond speculation. Indeed, railroads needed a lot of upfront investment and produced little or no return in the early years, presenting the risk to investors.

Panic of 1873

Rapid growth of railroad investing
around the world

Causes of panic:

- Boston and Chicago fires
- Abandon silver currencies
- Franco-German war



There were a number of contributing factors that eventually led to the burst of the railroad speculation bubble. For instance, the Boston and Chicago fires right before the crisis reduced bank reserves. Germany and the US abandoned silver currencies, which put strains on miners and reduced the need for railroad networks. And dislocations from the Franco-German war were still present. The first signs of the crisis became visible in Vienna, where the stock exchange crashed on May 9th, 1873, and a series of bankruptcies followed. In Germany, a large railway company collapsed, adding to the panic. In the US, J Koch and Company, a major bank, was unable to sell railway bonds. It collapsed mid-September, sparking bank runs and failures across the US.



A prominent victim of the US banking crises was the Friedman Savings Bank. This was the first and only bank at the time that served the free former slaves.

Panic of 1873

Rapid growth of railroad investing
around the world

Causes of panic:

- Boston and Chicago fires

- Abandon silver currencies

- Franco-German war

*Common exposure to speculative
investment*



The US and the Europe plunged into a deep recession with unemployment rising fast. This crisis also affected the Ottoman Empire and Latin America. All told, this crisis originated in speculative investments, real world bonds that led to large losses, and a lack of trust in the banking system.

Barings Crisis of 1890



A different type of investment, namely government debt, sparked the so-called Barings Crisis of 1890. By 1890, Argentina had experienced a decade of large capital inflows.



The capital was raised mostly in London by European banks, including Barings bank.

Barings Crisis of 1890

Long-term infrastructure investment

Causes of panic:

- Excessive debt

- Collapse of bond issuance

- Government default

Speculation by international investors



The money was used to finance long-term investment projects in infrastructure, including the building of railroads and transportation networks, and the improvement of farmlands. However, by the late 1880s, as much as 40% of foreign borrowing was going to debt service and 60% of imports was going towards consumption goods. The Argentinian government was also running a substantial budget deficit. Railway net profits had peaked in 1888 and eventually international investors became wary of Argentina's government's debt burden. Eventually, Barings could not find investors for Argentina's debt issuance. Argentina experienced bank runs and unable to raise new funds, defaulted on its government bonds. This default led to a sharp recession in Latin America, but also to sharp stock market drops in Europe and the US. As with the 1873 Crisis, a credit boom to speculative long-term investments eventually turned bad. But note the international linkages. The speculative activity was fueled by international investors.

Panic of 1907

Large railroad and insurance losses

Copper speculation financed by credit

Speculation by international investors



The next crisis, the Panic of 1907, originated in copper speculation. Since we discussed the Panic of 1907 in a separate lecture, I will be brief. The San Francisco earthquake caused large losses for insurance companies, railways, and banks. Against this backdrop, speculation in copper equities gone wrong led to even more losses in the banking system. This led to a banking panic, bank closures, and a recession that spread from the US to Europe. In the US, the unemployment rate rose to over 16%. Global markets were also hit hard. For instance, the British unemployment rate doubled from 3.6% to 7.8% between 1906 and 1908. Germany, Japan, and Italy also experienced bank runs. The Panic of 1907 is an example of financial market speculation gone wrong. However, the effects are like another crisis. Bank losses caused bank runs. Then the disruption in the banking system caused credit and the real economy to contract. Here too, there were international linkages. This included overseas exposures to insurance companies, underwriting disaster risk in San Francisco, and risky stock market bets by foreign investors. This allowed the crisis to spread from the US to the rest of the world.

Great Depression of 1930s

Gold standard

Domestic currency convertible to gold



The Great Depression provides the clearest example of the importance of international linkages. Since we discussed the Great Depression separately, let me just focus on the international component. In 1929, when the stock market crash in the US triggered the Great Depression, most countries were on the gold standard. The gold standard meant that all paper currency was convertible into gold and gold reserve therefore dictated the money supply.



This also meant that most countries operated at a fixed exchange rate system as each currency was fixed to gold.

Great Depression of 1930s

Gold standard

Domestic currency convertible to gold

Gold reserves $\Rightarrow$ money supply

Fixed exchange rate system

U.S. hoards gold $\Rightarrow$ international shock

Fixed exchange rate creates spillovers



At that time, the US was a net lender of gold in the global markets. However, when the banking crisis unfolded, the US hoarded gold rather than lending it out. This left other countries scrambling to maintain their gold reserves to avoid a decrease in their respective money supply. A lower money supply puts strain on banks and led to

deflation around the world. For instance, the German Reichsbank basically went out of gold when short term creditors stopped lending to Germany. In the end, the gold standard was abandoned to allow countries to control their own money supply and avoid bank runs and deflation. In a nutshell, the Great Depression was transmitted through the global financial system because of fixed exchange rates. So different from earlier financial crisis, the issue here was not a common exposure to a bad long-term investment. This time, all currencies and all money supplies around the world were linked. With floating exchange rates, the value of the currency adjusts when facing an international shock. With fixed exchange rate, the money supply must adjust with large negative consequences.

The slide features a dark blue background. On the right side, there is a video frame of a man in a dark suit and light blue shirt, gesturing with his hands as if speaking. In the top right corner of the slide, there is a small red 'I' logo. The main title 'Financial Crisis of 2008' is in large white font. Below it, the text 'This time is the same again:' is followed by two bullet points: 'Long-term investment (in housing)' and 'Financed by short-term credit', both in white font.

Financial Crisis of 2008

This time is the same again:

- Long-term investment (in housing)
- Financed by short-term credit

That brings us all the way to the 2008 Financial Crisis. You might now see how similar this event was to past crises. The crisis originated in the US housing market. Also, many other countries. Such as Spain and Ireland had also experienced a housing boom and bust. Many long-term mortgage investments funding the housing boom were financed by short-term credit. Once this short-term credit dried up, the mortgage market and house prices collapsed. As in previous financial crisis, international investors had gained exposure to these investments. The extent to this exposure became clear when a mid-sized German bank, the ECA Bay Deutsche Industriebank, failed in June 2007 due to losses from US subprime mortgages. In addition, it became apparent that domestic players, most notoriously the investment bank Lehman Brothers, depended on short-term funding secured by mortgages.

Lehman Brothers



Lehman Brothers: © 2007 Shankbone, D. / CC BY-SA 3.0 / Wikimedia / Lehman Brothers Times Square

American International Group



Lehman Brothers collapsed in fall of 2008. So did the largest insurance company, the American International Group. This triggered a stock market crash and panic in the funding markets worldwide.

Financial Crisis of 2008

This time is the same again:

Long-term investment (in housing)

Financed by short-term credit

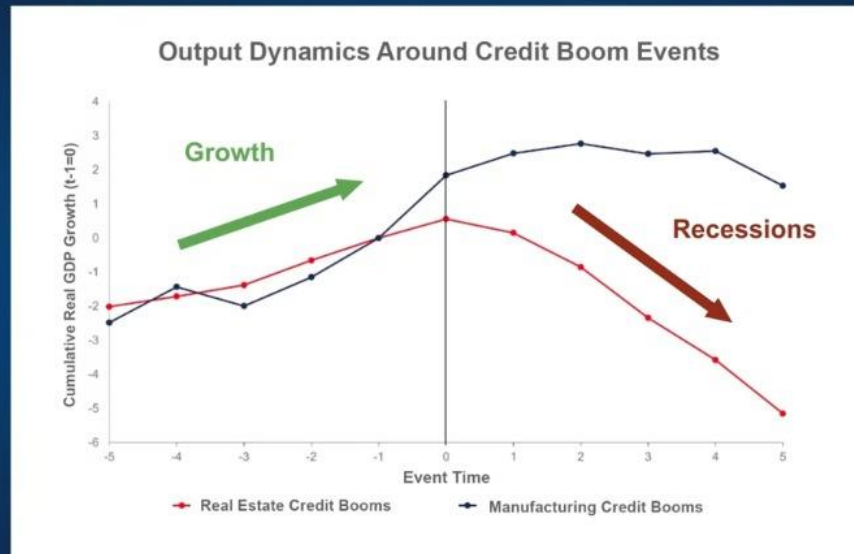
Common exposure to speculative investment

Globally integrated financial system



In 2008, the Crisis unfolded worldwide because of the globally integrated financial system in which capital flows freely across borders. A globally integrated financial

system helps when shocks are small as they can be absorbed easily. However, a shock to the largest economy impacts everyone.



Before we wrap up, let's consider one final question. Do credit booms always lead to financial crises? The short answer is no. Recent research shows that many credit booms do not result in financial crises, but instead can have sustained economic growth. This chart shows output dynamics before and after the peak of a credit boom for a sample of 189 countries from 1910 to 2014. The peak of the credit boom is indicated by event time equal to zero, the vertical line. There are two lines indicating two different types of credit boom. Roughly speaking, one is in construction and real estate, the other is in the manufacturing sector. What does this graph show? Well, on average, GDP grows strongly during the boom. However, after the peak, output only seems to decline after a real estate credit boom.

Credit Booms Aren't Always Bad

189 countries from 1910—2014

Real estate credit booms

Limited by local demand

Manufacturing credit booms

Global demand ⇒ sustainable



Why does this happen? Typically, a real estate credit boom increases house prices, construction of new homes, employment, and so on. However, eventually there will not be enough local buyers to sustain the boom. This will then put downwards pressure on house prices, which can lead to a collapse in the construction sector. In contrast, credit booms in the manufacturing sector can be more sustainable. This usually happens when they are driven by global demand or demand for new products. Think about the fast increase in manufacturing in South Korea from the 1960s to the 1980s. Most of these products were exported. Credit to the manufacturing sector can fund profitable investments. These higher profits help to pay off the debt incurred during the credit boom.

Summary

Common features of financial crises

- Excessive (short-term) debt
- Bad long-term investments
- Unsustainable government policy
- International linkages

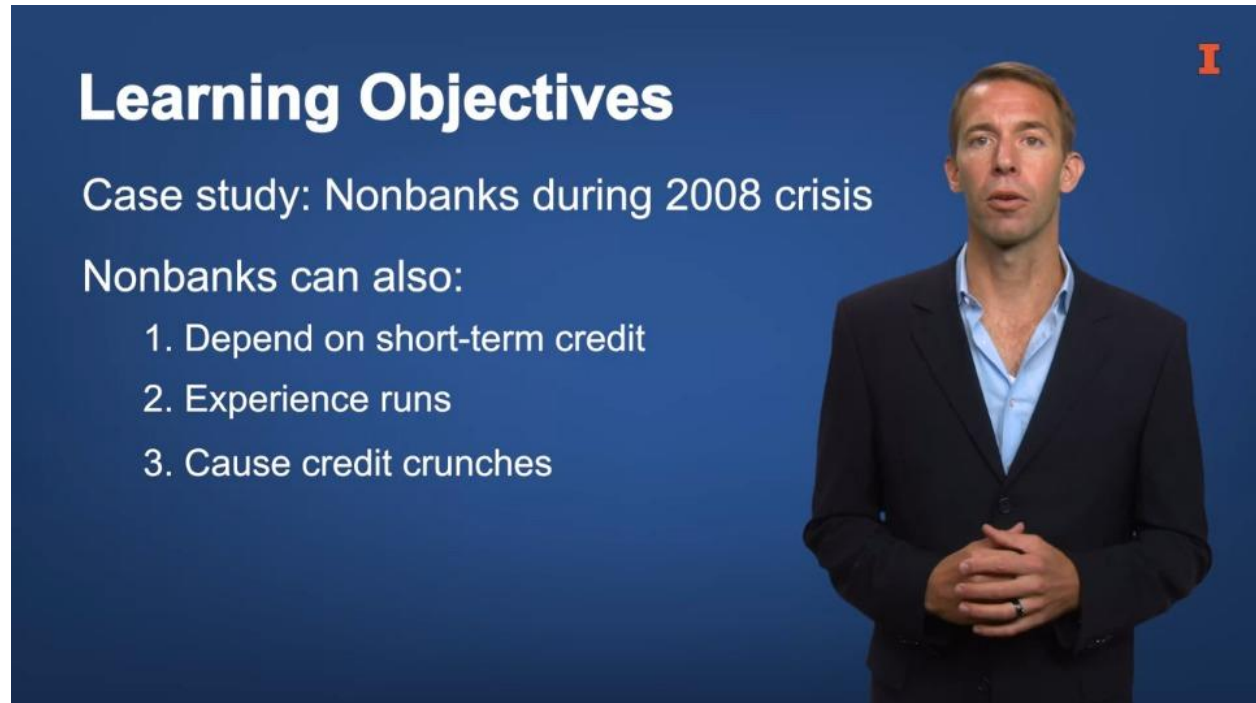
Not all credit booms are bad!



Let's wrap up. In this session, we have described the common features of financial crisis. They include excessive debt, especially short-term debt, long-term investments that eventually go bad, or some unsustainable government policy such as excessive spending or fixed exchange rates. We also saw how some financial crisis became global in nature. Last, we saw that that not all credit booms lead to banking crises or drops in GDP. Historically, credit booms in the manufacturing sector have been more sustainable.

Lesson 4-1: Understanding Financial Crises

Lesson 4-1.4. Auto Loans During the 2008 Financial Crisis



Learning Objectives

Case study: Nonbanks during 2008 crisis

Nonbanks can also:

1. Depend on short-term credit
2. Experience runs
3. Cause credit crunches

Hello and welcome to this lecture on auto loans during the 2008 financial crisis. In this case study, we will look at what happened in the auto loan market. A market in which non-bank lenders play a large role when short term funding dried up. We will see that not only banks are susceptible to runs, and that a drop in credit supply led to a drop in auto sales.

Captive Finance Companies

Buy a car from Ford, get a loan from Ford

50% of U.S. auto loan market in 2008

Lender affiliated with auto maker

Ford ⇒ Ford Motor Credit

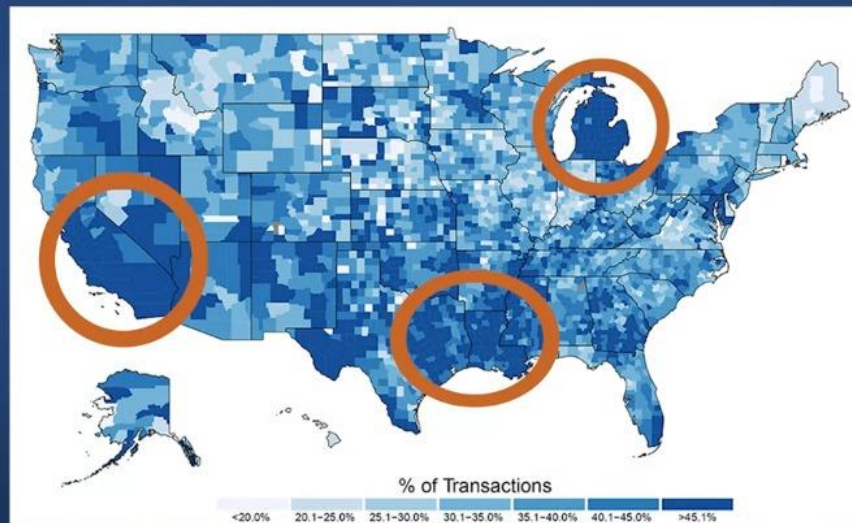
General Motors ⇒ GM Financial

Toyota ⇒ Toyota Financial Services



Before the 2008 financial crisis, about half of all auto loans were originated by captive finance companies. A captive finance company is a lender that is owned by the automaker, for instance, Ford Motor Credit, GM Financial, or Toyota Financial Services.

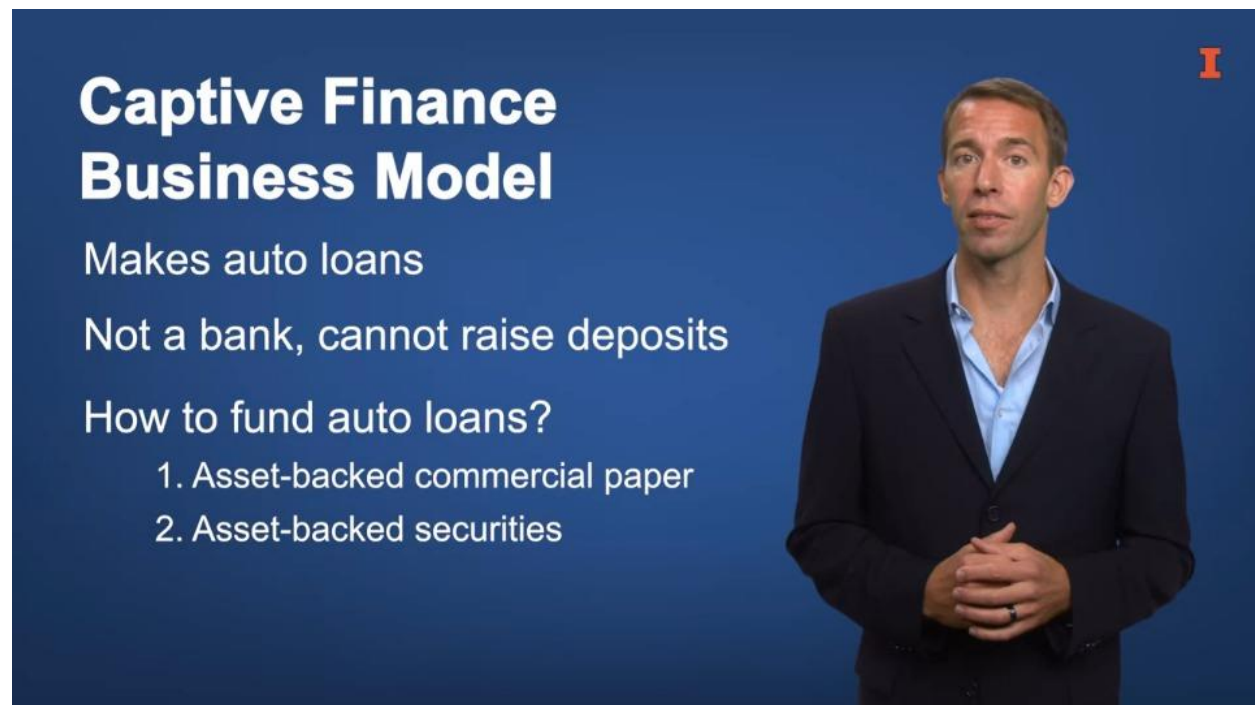
Share of Captive Finance in Auto Loans (2008:Q1)



IHS. Polk's New Vehicle Registration Data. <https://www.ihs.com/btp/polk.html>

The map shows you the distribution of market share of captive finance companies across US counties. As you can see, the market share is particularly high in Michigan where most of the automakers are located. In the South, which historically had less

banks than the North and in the West. In other words, captive finance companies were particularly strong where there was less access to bank credit.

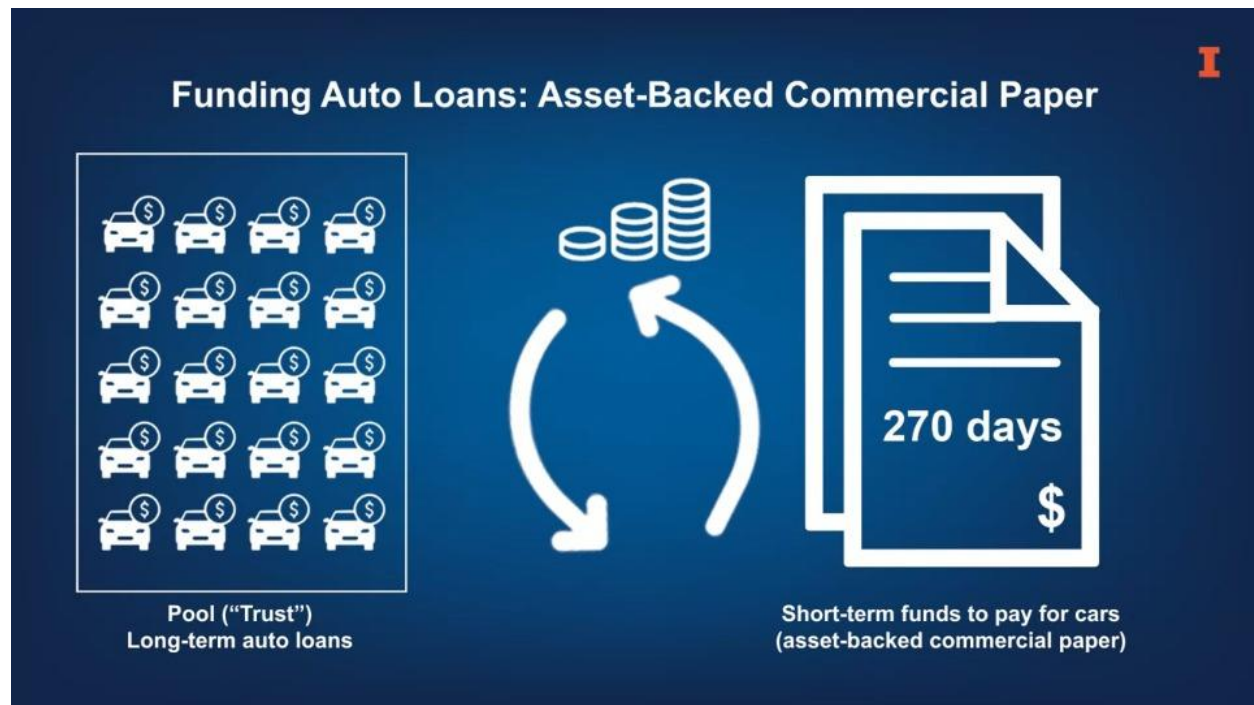


Captive Finance Business Model

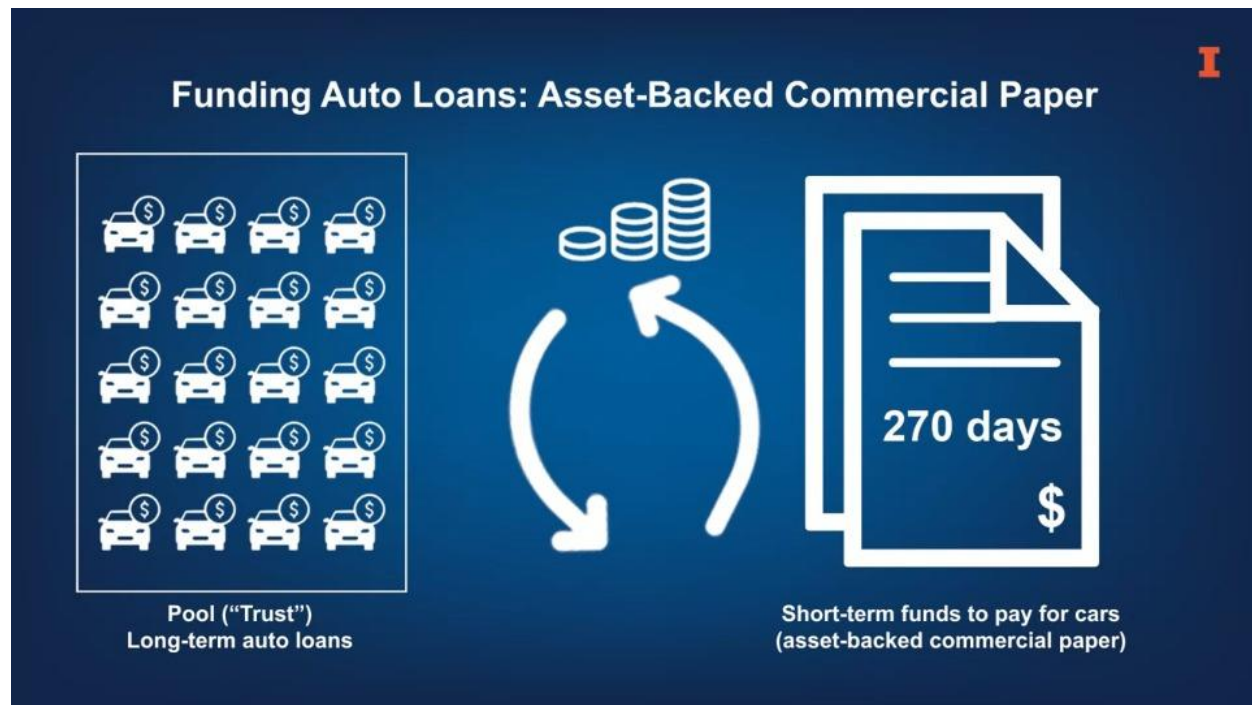
- Makes auto loans
- Not a bank, cannot raise deposits
- How to fund auto loans?
 1. Asset-backed commercial paper
 2. Asset-backed securities

The slide features a man in a dark suit and light blue shirt standing on the right side, with his hands clasped. The background is a solid blue color. The text is white and bold.

Different from banks, captive finance companies do not take deposits. Instead, captive finance companies funded themselves by issuing asset-backed commercial paper and asset-backed securities.



Let's first discuss how asset backed commercial paper worked in this context. A captive finance company would establish a trust and transfer auto loans into that trust. The trust would then issue commercial paper that is backed by the auto loans and the cash flow from these loans. The trust uses the proceeds of the asset backed commercial paper issuance to pay the captive finance company for the loans. Typically, these trusts are revolving. This means that the captive finance company puts in new order loans as old auto loans are being paid off. By doing so, the captive finance company has continuously access to short term funding which in turn allows the captive company to issue new loans. However, commercial paper has a maturity of at most 270 days and needs to be rolled over continuously. This means that the captive finance companies, just like banks, borrow short term and lend long term. That is, they engage in maturity transformation.



Asset-backed securities are slightly different. Here the captive finance company sets up a trust and transfers the auto loans into that trust. The trust raises money by issuing longer term notes.

I

\$999,980,000

Ford Credit Auto Owner Trust 2018-B

Issuing Entity or Trust
(CIK: 0001754784)

Ford Credit Auto Receivables Two LLC

Depositor
(CIK: 0001129987)

Ford Motor Credit Company LLC

Sponsor and Servicer
(CIK: 0000038009)

Before you purchase any notes, be sure you understand the structure and the risks. You should read carefully the risk factors beginning on page 17 of this prospectus.

The notes will be obligations of the issuing entity only and will not be obligations of or interests in the sponsor, the depositor or any of their affiliates.

The trust will issue:	Principal Amount	Interest Rate	Final Scheduled Payment Date
Class A-1 notes	\$ 205,000,000	2.50308%	November 15, 2019
Class A-2a notes	275,000,000	2.96%	September 15, 2021
Class A-2b notes ⁽¹⁾⁽²⁾	100,000,000	one-month LIBOR + 0.12%	September 15, 2021
Class A-3 notes	317,000,000	3.24%	April 15, 2023
Class A-4 notes	102,980,000	3.38%	March 15, 2024
Class B notes ⁽³⁾	31,580,000	3.53%	May 15, 2024
Class C notes ⁽³⁾	21,050,000	3.68%	April 15, 2025
Total	\$ 1,052,610,000		

Here's an example from Ford Motor Credit Owner Trust 2018-B that issues almost one billion in notes. The notes will be backed by a pool of car, light truck, and utility vehicle loans bought by Ford Motor Credit. As you can see in this transaction, the maximum

maturity is seven years, a much longer financing horizon. There is not much maturity transformation in this deal. At the same time, you can also see that the longer maturity, the higher the interest rate on the note. Moreover, this is a one-time deal, meaning that different from asset-backed commercial paper, the pool of loans will not be replenished, and the loan payments are used to pay off the notes. Before the 2008 financial crisis, asset-backed commercial paper was the main source of funding for captive finance companies because it was cheaper than asset-backed securities and revolving.



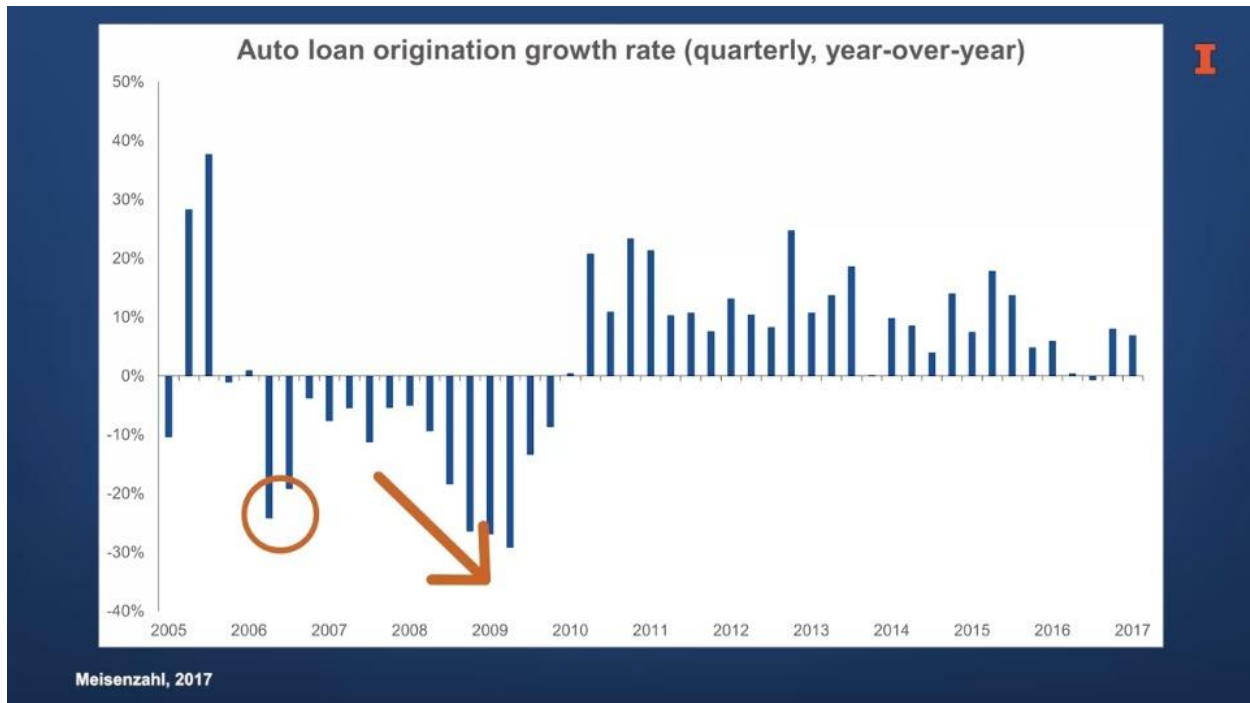
2008 Financial Crisis

- Concerns about potential loan losses
- Investors withdrew short-term funding
- Finance company response (e.g., GMAC)
 - Less funding for auto loans
 - Tighten credit standards

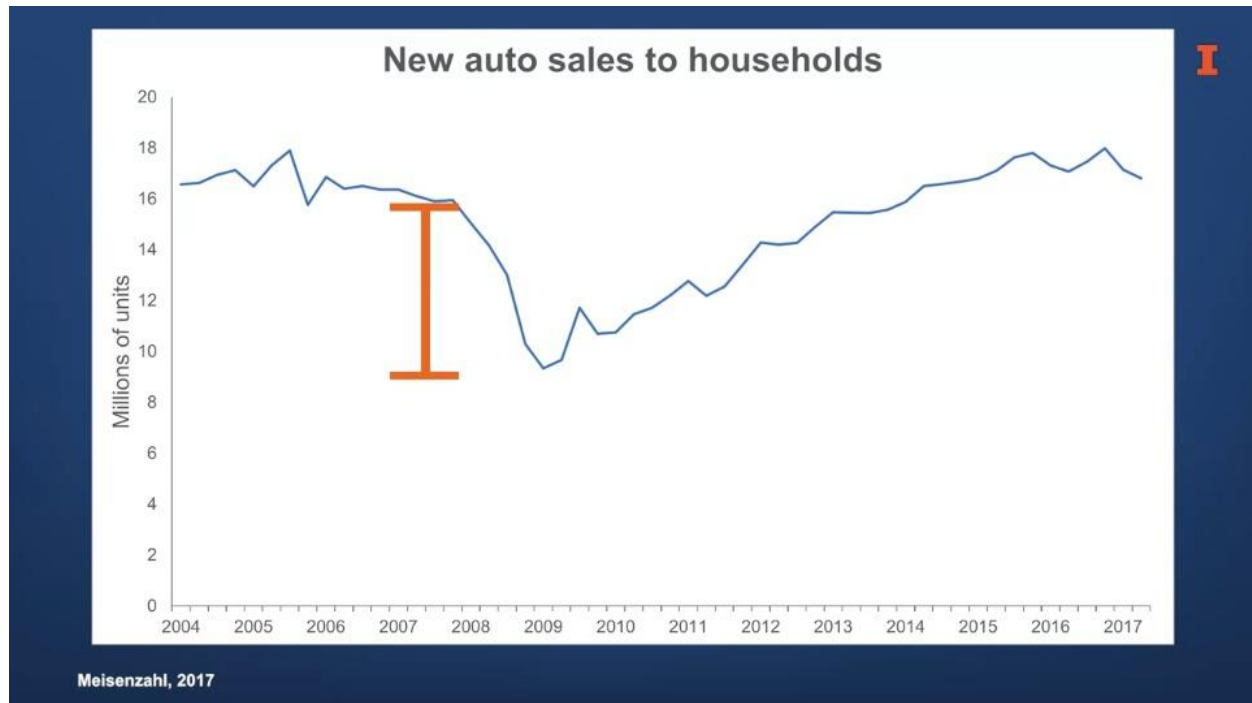
The slide features a blue background with white text. On the right side, there is a photograph of a man in a dark suit and light blue shirt, standing and gesturing with his hands. A small red 'I' logo is visible in the top right corner of the slide.

However, when the subprime crisis started to hit the mortgage market in 2006 and 2007, investors became suspicious of the qualities of loan pools backing asset-backed commercial paper. One question that was asked was how many subprime auto loans were in those loan pools. When investors discovered that the pool from the General Motors Acceptance Corporation was of low quality, the concern spilled over into the auto loan market. Investors were less and less willing to roll over asset-backed commercial paper backed by auto loans, reducing funding available to captive finance companies. The response of General Motors Acceptance Corporation was dramatic. In October 2008, General Motors Acceptance Corporation announced that it would considerably tighten credit standards and would only lend to consumers with a credit score above 700. At the same time, the median credit score was 720. Meaning a large number of potential auto buyers would be denied credit. Other captive finance companies had also already tightened their credit standards. Nevertheless, investors refused to roll over asset back commercial paper, and this funding dried up completely. The collapse of the trust meant that the captive finance companies had to pay off the asset back commercial paper. General Motors Acceptance Corporation and Chrysler

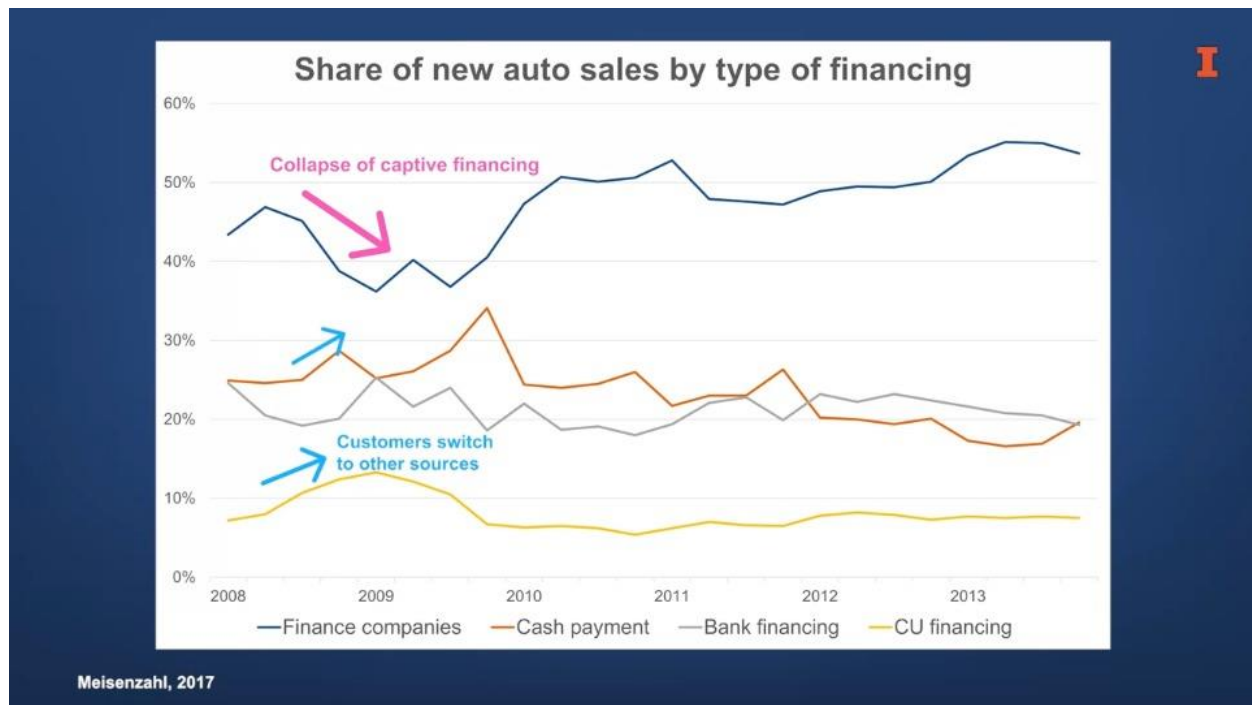
Financial Incorporated were unable to meet their obligations and collapsed. The unwillingness of investors to roll over the asset-backed commercial paper is in essence the same as a bank run. Short term funding is withdrawn by all investors simultaneously leading to the collapse and liquidation of the trust. Let's look what this run on the asset backed commercial paper trust and the resulting credit crunch meant for auto loans.



This graph shows you the year over year growth rate of auto loans. As you can see, they turned negative in early 2006 when the subprime crisis in the mortgage market started to unfold. But when investors stopped rolling over asset backed commercial paper in mid 2008, loan originations dropped by over 30 percent when compared to the previous year.



What happened to auto sales? As you can see in this graph, before funding for finance corporations dried up, US households bought about 16 million new cars at an annual rate. When the financial crisis hit, retail auto sales collapsed to 7.9 million at an annual rate in the first quarter of 2009. The timing suggests that the drop in auto sales was driven by a drop in credit supply. But how can we be sure that the drop in auto sales was not driven by low demand? During the financial crisis, many workers lost their job, and bleak economic prospects could have made consumers hesitant to buy new cars. If the collapse was driven by demand and not by reduced credit supply, we should see that other lenders were not stepping in and making loans when captive finance companies could not.



However, this graph shows you that in 2008, more auto buyers paid either with cash or got credit from a bank or a credit union. This suggests that runs on asset backed commercial paper had significant consequences for auto sales. The example of captive finance companies illustrates that runs can occur outside the banking sector. Indeed, every financial institution that engages in maturity transformation can be subject to a run.

Features of (Some) Nonbanks

- Depend on short-term debt
- Experience runs
- Contribute to credit collapses
- Before 2008, ignored by bank regulators
- No access to lender of last resort (Fed)



However, before the 2008 financial crisis, regulators had largely ignored maturity transformation and credit provision by nonbank financial institutions. These nonbank financial institutions had no access to the Federal Reserve as the lender of last resort, that would help to stop the runs in the short-term funding markets. These runs, through their effects on credit supply, contributed to the depth of the recession that followed.

Summary

- Case study: Captive finance companies
- Like banks, captives borrow short and lend long
- Like banks, captives experience runs
- Runs reduced credit and auto sales

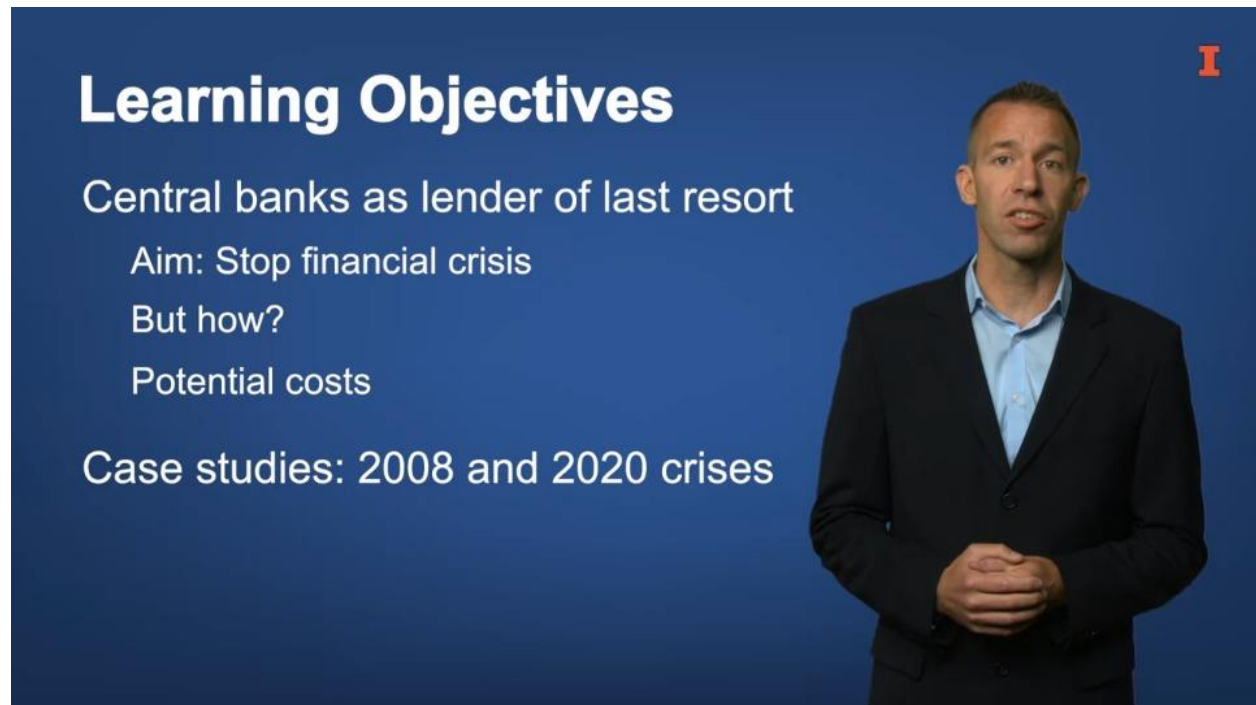


What have we learned in this lesson? First, with captive finance companies as an

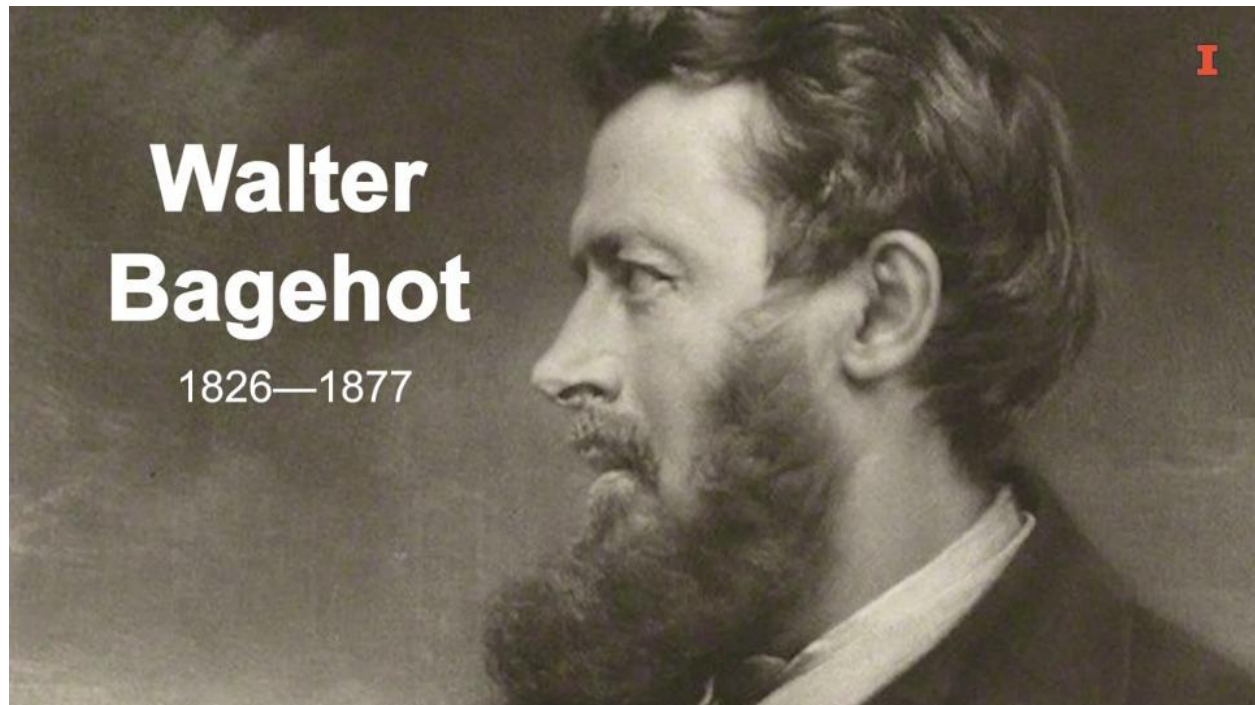
example, we showed that non-bank financial institutions engage in maturity transformation. Second, non-bank financial institutions experienced runs in the funding markets during the 2008 financial crisis. Third, the runs reduced auto credit and therefore auto sales.

Lesson 4-2: Policy Responses to Financial Crises

Lesson 4-2.1. The Fed as Lender of Last Resort

A video frame featuring a man in a dark suit and light blue shirt standing against a solid blue background. He is positioned on the right side of the frame, looking towards the camera with his hands clasped in front of him. On the left side of the frame, there is white text. In the top right corner of the blue area, there is a small red 'I' logo. The text on the left reads: 'Learning Objectives' in a large, bold font, followed by 'Central banks as lender of last resort' in a slightly smaller font. Below this, there are three bullet points: 'Aim: Stop financial crisis', 'But how?', and 'Potential costs'. At the bottom of the text block, it says 'Case studies: 2008 and 2020 crises'.

Hello and welcome to this class on the lender of last resort function of central banks. The Panic of 1907, the Great Depression, the global financial crisis of 2008. Banking panics and financial crises can have devastating impacts on the financial sector and ultimately the real economy. In this session, we will examine how a central bank, by providing short term cash loans during a financial crisis, can help to stabilize financial markets and prevent the worst-case scenario. We will see how these interventions should be designed in principle, but also how they work in practice. We will also understand the potential costs of these interventions. We will examine these issues in the context of the 2008 financial crisis and the 2020 global pandemic.



Let's get started. The idea that a government backed central bank should intervene and stop a financial crisis date to Walter Bagehot, a journalist from the 19th century England.

Bagehot's advice in 1873

“In times of financial crisis, **central banks should lend freely to solvent depository institutions**, yet only against sound collateral and at interest rates high enough to dissuade those borrowers that are not genuinely in need.”

In 1873, Bagehot famously said that in times of financial crisis, central banks should lend freely to solvent depository institutions, yet only against sound collateral and at interest rates high enough to dissuade those borrowers that are not genuinely in need.

This statement has become a touchstone for how central banks should respond to financial crises. Let's quickly unpack what it means. Let's start with “central banks should lend freely to solvent depository institutions”. Sometimes healthy banks are swept up in a banking panic. Since all banks borrow short and lend long, some of these healthy banks might run out of cash. This is who the central bank should support, the solvent banks.

Bagehot's advice in 1873

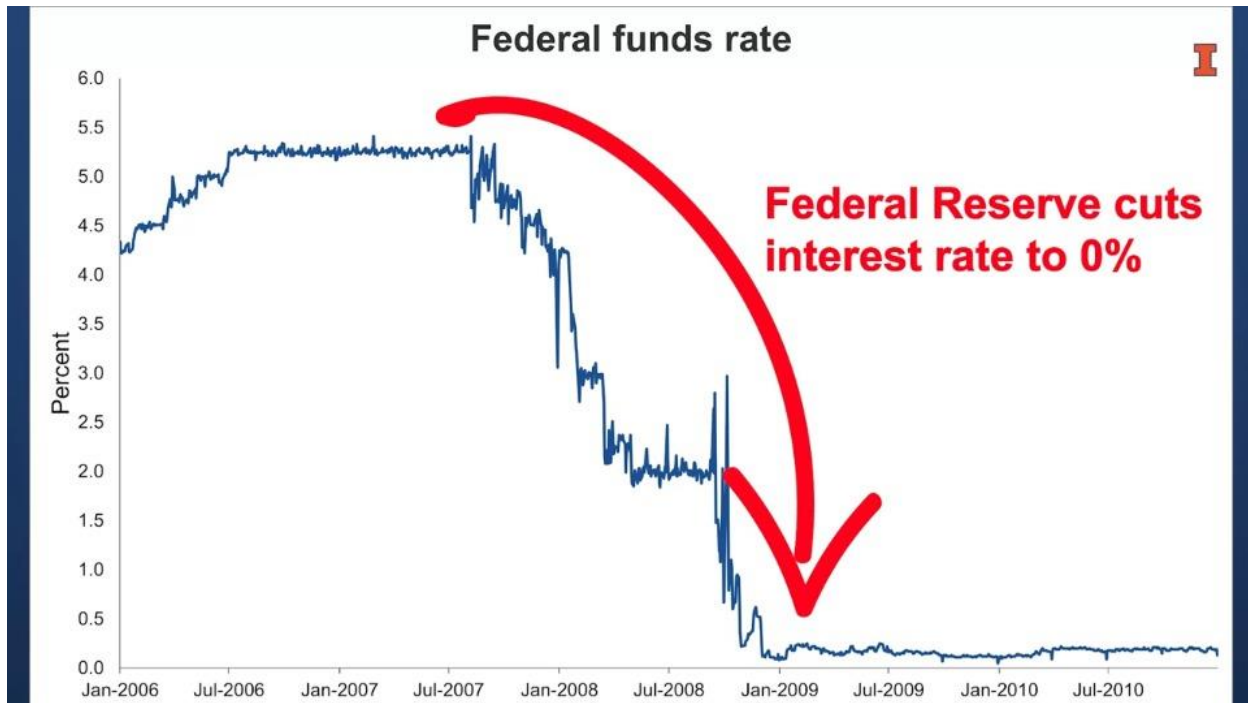
“In times of financial crisis, central banks should lend freely to solvent depository institutions, yet only **against sound collateral and at interest rates high enough to dissuade those borrowers that are not genuinely in need.**”

What about the second part? Against sound collateral and at interest rates high enough to dissuade those borrowers that are not genuinely in need. What does this achieve? First, the central bank should not lose money when it intervenes. These programs should not create losses for taxpayers. Second, borrowers should only tap the central bank when no other options are available. This should be a temporary support and not a long-term solution. The borrowers should want to pay back any central bank loans as soon as they can. Bagehot thought the government should intervene but do so with care. The ultimate goal is to stop the banking panic without losses to the taxpayer. That's the theory. Let's see how this works in practice. Let's consider central bank interventions in the two recent crises, the financial crisis of 2008, and the global pandemic of 2020.



Case Study 1: 2008 Financial Crisis

Let's begin with the 2008 episode. This was triggered by the contraction of the US housing market that began in 2006. The decline in house prices was accompanied by a rise in mortgage delinquencies. As losses mounted, there was a sharp reduction in credit supply, especially in short term funding markets. This cumulated with the collapse of Lehman Brothers, a well-known investment bank, which sent shockwaves through global financial markets. By the fall of 2008, the US was on the brink of a deep recession.



How did the Federal Reserve respond? Well, the first thing they did was to cut interest rates from 5 to 1.25% in mid 2007, to effectively 0%. In addition, the Federal Reserve set up a variety of emergency liquidity programs to stabilize financial institutions and credit markets. The Fed wanted to stop the panic after Lehman Brothers failed. This was the Fed acting as the lender of last resort.

Fed as LOLR in 2008

Goal: stabilize funding markets

How? Emergency cash loans

1. Banks needed longer-term funding
2. Frozen funding markets
3. Nonbanks needed cash



Traditionally, the Federal Reserve would have provided emergency cash loans to only

banks through the discount window. However, in 2008, this was not going to work. Banks were no longer the only game in town. Non-banks played a crucial role in many funding markets. And many of these non-banks were swept up in the panic. For example, Lehman Brothers was an investment bank. It did not have access to the discount window. To stabilize funding markets, the Federal Reserve acted decisively by extending the reach of the discount window along three dimensions. First, longer term funding tried up for banks. The Federal Reserve established the Term Auction Facility or TAF. Second, securitization markets had collapsed after the bankruptcy of Lehman Brothers. The Federal Reserve established the Term Asset-backed-securities Loan Facility, or TALF to unfreeze this market. Participants in the TALF could get long term loans directly from the Fed to buy securitized loans. Third, many non-banks that were central to funding markets desperately needed cash. The most important and reputable investment banks are also called primary dealers. Primary dealers serve as the trading counterparties for the federal reserves, open market operations. They are crucial for monetary policy. They are also vital for well-functioning treasury markets.

Fed as LOLR in 2008

Goal: stabilize funding markets

How? Emergency cash loans

1. Banks needed longer-term funding

2. Frozen funding markets

3. Nonbanks needed cash

Primary Dealer Credit Facility (PDCF)

Term Securities Lending Facility (TSLF)

Asset-Backed Commercial Paper Money Market Mutual Fund
Liquidity Facility (AMLF)

Commercial Paper Funding Facility (CPFF)

The Primary Dealer Credit Facility, or PDCF, and Term Securities Lending Facility, or TSLF, were established to meet their needs. Money market funds were experience runs, so the Fed established the Asset-Backed Commercial Paper Money Market Mutual Fund Liquidity Facility, or AMLF. Finally, the Commercial Paper Funding Facility, or CPFF, provided support for US issuers of commercial paper. This is the only example of a direct purchase program in 2008. This meant that the Federal Reserve bought commercial paper out right. While this was mostly issued by financial institutions, the

Fed did end up buying paper issued by non-financials, such as Harley Davidson or McDonald's. By historical standards, this was a highly unusual move.



Was the LOLR Successful?

- Stopped panic
- Borrowers quickly repaid loans
- Zero losses
- But ... unprecedented expansion of safety net

That was a large and complex intervention. Was it successful? Would vulture budget be happy? Well, first and foremost, the Federal Reserve did stop the financial panic and in each of these programs the emergency loans were backed by high quality collateral and above market interest rates were charged. The loans were repaid quickly, and the programs were wound down without any losses. On the other hand, this was a massive and unprecedented expansion of the government safety net. Many of the financial institutions receiving support were not depository institutions. They were not supervised or regulated by the Federal Reserve or another bank regulator. What if market participants continue to expect the Federal Reserve to offer support when things go back? How will this affect risk taking? There is no clear answer to this question.



Case Study 2: 2020 Pandemic

Let's fast forward to 2020. The global pandemic arrived in the US in the first quarter of 2020. By March 2020, there was severe volatility in the treasury market which was likely to cause disruptions in short term funding markets. This set off alarm bells around the world. Since the 2008 financial crisis showed how bad it can get, policymakers rushed to provide first support to the treasury market and then more broadly. As before, this included the Federal Reserve acting as the lender of last resort.

Fed as LOLR in 2020

Same playbook as 2008

Paycheck Protection Program
Liquidity Facility (PPPLF)

Direct purchase programs:

Primary Market Corporate Credit Facility
(PMCCF)

Secondary Market Corporate Credit Facility
(PMCCF)

Main Street Lending Program



While the shock was very different, we will see the Central Bank's response was similar. Several liquidity facilities used during the 2008 financial crisis were rolled out again, including the Term-Asset-Backed Securities Loan Facility, TALF, the Commercial Paper Funding Facility, CPFF, and the Money Market Mutual Fund Liquidity Facility, MMLF. The perhaps largest program was the Paycheck Protection Program Liquidity Facility, or PPPLF. The Paycheck Protection Program was a government program providing 600 billion in forgivable loans to small businesses. Under the PPPLF, lenders could receive funding for these loans from the Central Bank. The direct purchase programs played a much bigger role this time around. The Primary and Secondary Market Corporate Credit Facilities, the PSMCCF provided direct support to large businesses through the corporate bond market. The Fed was directly purchasing corporate bonds on a large scale. By April 2020, the Federal Reserve was also involved in direct lending to small businesses. The main street lending program allowed the Fed to partner with banks to originate small business loans.

Lessons from LOLR in 2020

Combination of fiscal and monetary policy

Support for nonbanks inevitable?

Direct purchase programs inevitable?



Where does this leave us? What more can we learn about the lender of last resort from the 2020 episode? Well, let's start with the good news. First, these interventions look successful in avoiding a total meltdown of the economy. Second, we learn that crisis interventions seem to work best when fiscal and monetary policy work hand in hand. Perhaps the most successful government intervention was the Paycheck Protection Program. We can easily see why the Federal Reserve providing the 600 billion funding for these loans was crucial during a time of crisis. At this point, banks probably would have rather hoarded cash than make additional loans. How would Walter Bechet feel about all of this? Well, he only thought central banks should support depository institutions. That seems to be a thing of the past. He was also concerned about potential losses to taxpayers. Well, unfortunately, going from 2008 to 2020 we saw a far greater involvement of the Federal Reserve in direct lending to small and large businesses. This of course means that the Federal Reserve is directly exposed to losses if defaults in the corporate sector were to occur. That would surely make [inaudible] feel uneasy.

Summary

Lender of last resort ⇒ stop crisis

2008 and 2020 case studies

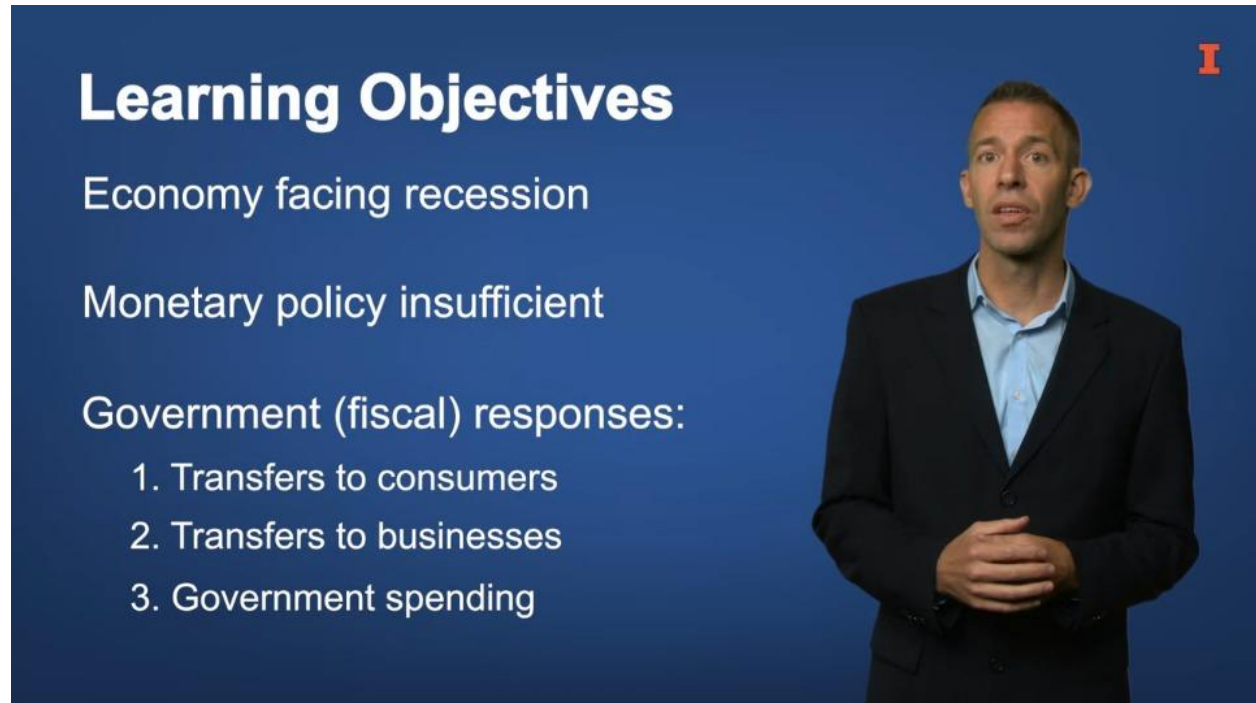
Potential issues in recent years



Let's wrap up. What have we learned in this session? The lender of last resort can stop financial crisis through emergency lending programs. Historically, these funds flowed to banks and loans required high quality collateral and above market interest rates. We saw how the Federal Reserve intervened to calm markets in 2008 and 2020. Finally, we discussed how the footprint of these emergency lending programs has expanded in recent years to cover non-banks as well as direct lending programs.

Lesson 4-2: Policy Responses to Financial Crises

[Lesson 4-2.2. Fiscal Responses to Financial Crises](#)



Learning Objectives

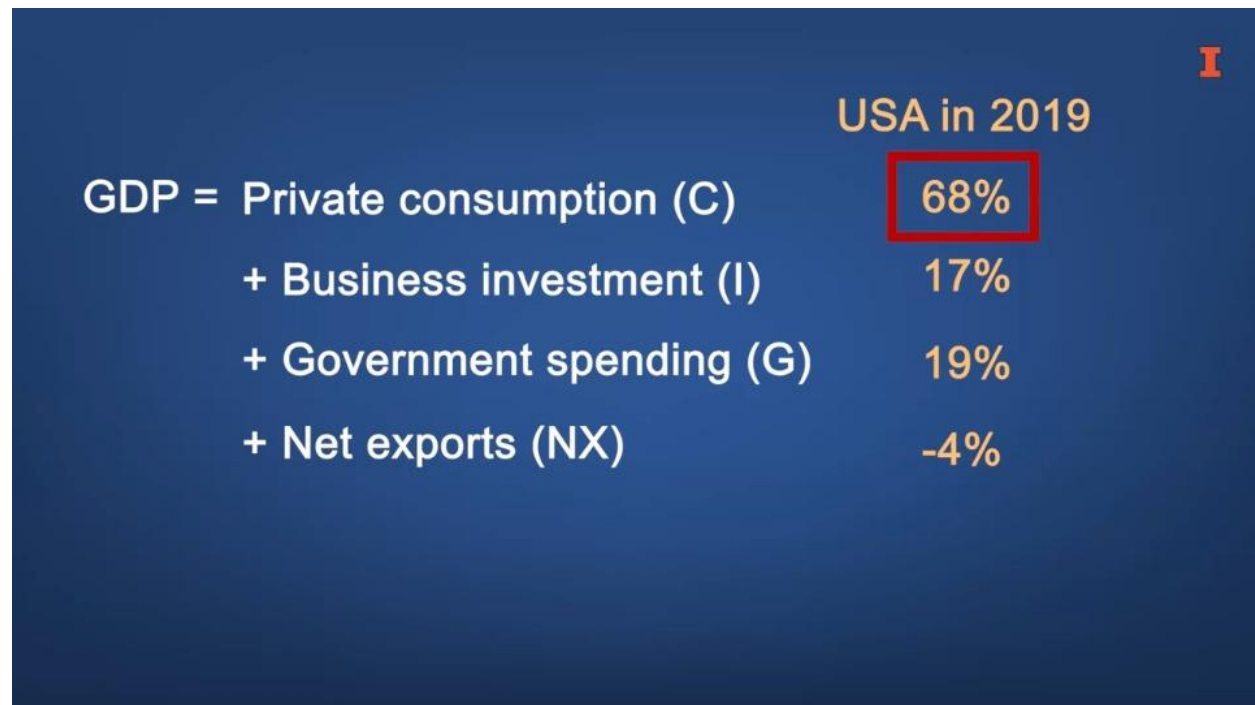
Economy facing recession

Monetary policy insufficient

Government (fiscal) responses:

1. Transfers to consumers
2. Transfers to businesses
3. Government spending

Hello and welcome to this class on the fiscal responses to financial crisis. In this session, we will discuss how fiscal policy can be used to reduce the harmful effects of a financial crisis and the recession that typically follows. Monetary policy alone cannot stimulate an economy that is facing a recession. Instead, we usually see the government deploy various interventions to aid the recovery. We will study three types of fiscal responses. First, government transfers to consumers. These include tax cuts or stimulus checks. Second, government support for businesses; and third, we will discuss direct spending by the government on items such as infrastructure projects. We will describe these approaches in turn and discuss their effectiveness. As before, we will consider the 2008 financial crisis and the 2020 global pandemic as case studies.



USA in 2019	
GDP = Private consumption (C)	68%
+ Business investment (I)	17%
+ Government spending (G)	19%
+ Net exports (NX)	-4%

Let's get started, in any economy, total output, or GDP, always consists of the following private consumption, C, business investment, I, and government spending, G, and net exports, which is exports sold overseas minus imports or NX. For the United States in 2019, consumption was about 68% of GDP. Business investment was 17%, government spending was 19%, and net exports was -4% which means the US imported more than it exported. So, consumption was by far the largest component of GDP at that time.

Private consumption collapses during recessions



U.S. Bureau of Economic Analysis, n.d.

Now, let's have a look at what happened to private consumption in bad times. We can see that during normal times, consumption grows steadily, but then it collapses during bad times, during the 2008 financial crisis and its aftermath and then there was a much sharper drop during the 2020 global pandemic. In both cases, unemployment soared.

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↓ GDP = <u>Private consumption (C)</u> ↓	68%
+ Business investment (I)	17%
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This reduced income available to households for consumption because consumption

accounts for two thirds of GDP. This had a massive impact on the economy. Naturally, fiscal responses to financial crises focus on propping up consumption.

1. Transfers to Consumers

Send all households money

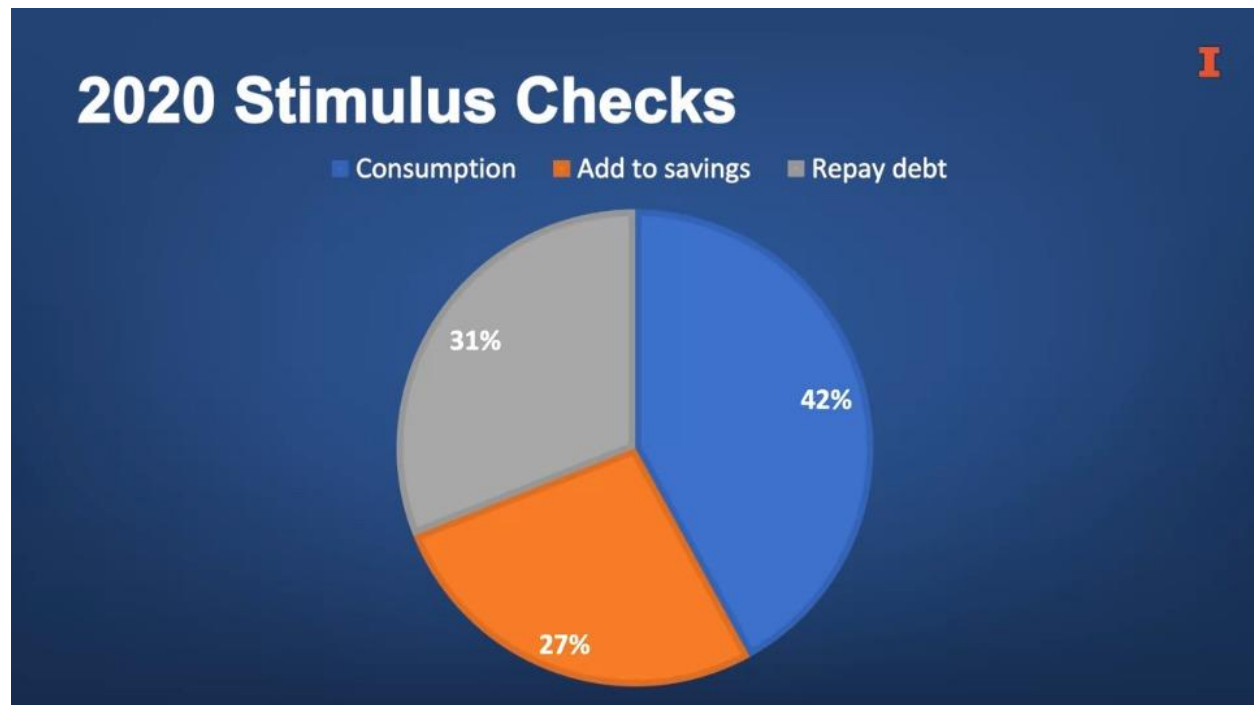
“Stimulus checks”

2008: up to \$1,200

2020: up to \$2,400



How does the government do this in practice? First, note that the newly unemployed reduce consumption the most, so cutting income taxes won't help as the newly unemployed do not have any income anymore. The other approach is to send out stimulus checks to all households. This has been done twice in the US in 2008, an individual taxpayer could receive a payment of up to \$600 and a couple up to \$1,200. In 2020, an individual taxpayer could receive up to \$1,200 and a couple up to \$2,400.



Are these stimulus checks effective? Well, yes and no. Recent research on the 2020 stimulus checks suggests that only 42% was spent on consumption, 27% was saved, and 31% was used to pay off debt, why wasn't all of this spent? People with lower incomes did spend more of their stimulus checks, however, people with high incomes saved it. Also, given the US economy was shut down, there were limited opportunities to spend the money.

1. Transfers to Consumers

Targeted interventions

Increase unemployment benefits

Government pays wage bill

Paycheck Protection Program



A second type of direct payment is more targeted. Increased unemployment benefits. Unemployment benefits were extended in 2009 and then increased and extended during the Covid-19 pandemic. This is a much more targeted interventions. Payments only go to people who recently lost their jobs. These households usually cut consumption most. Another approach is to pay businesses, not to lay off their employees. The government can cover part of the employer's wage bill. This approach was first tried out in Germany during the 2008 financial crises. The major advantage of this program is that we keep the match between the employer and the worker intact. This means the employee's skills do not depreciate. It also means that businesses do not need to rehire when demand increases again. In the United States during the Covid-19 pandemic, the Paycheck Protection Program had a similar intent.



The program offered loans to small businesses that would be forgiven if they did not fire their employees. In other words, the program was designed to ensure that employees were paid even if sales dried up due to the lockdowns. It also means that these employees retained their incomes and did not have to cut consumption. Putting these pieces together, direct transfers to households can support consumption and GDP during a recession. Programs targeted to the newly unemployed and programs keeping people employed are the most effective.

Investment share of gross domestic product (GDP) I



Moving on. The second component of GDP is investment. In this graph, we can see how much investment contributes to GDP. Investment collapsed during the 2008 financial crisis and fell once again in 2020.

2. Transfers to Businesses I

How to promote investment?

Tax breaks and subsidies



To counteract such drops, the government has limited options. The government can enact tax breaks or provide subsidies to foster investment. However, in the end,

investment is going to be driven by the state of the economy. For example, business investment will be driven by the demand for goods.



This brings us to the last type of fiscal response. The government can increase its own spending for instance, the American Recovery and Reinvestment act of 2009 allocated \$275,000,000,000 in federal contracts, grants, and loans to create jobs.



And then in 2021, in the wake of the global pandemic, the Infrastructure Investment and Jobs Act was passed.

3. Government Spending

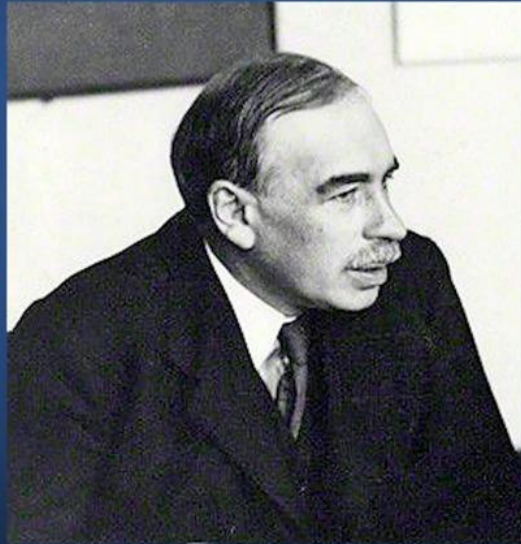
American Recovery and Reinvestment
Act of 2009

Infrastructure Investment and Jobs
Act of 2021



Both government spending bills included many public construction projects across the country. But is increasing government spending an effective tool to fight recessions?

John Maynard Keynes (1883—1946) **I**



The British economist John Maynard Keynes certainly thought so.

$$\begin{aligned} \text{GDP} = & \text{Private consumption (C)} & \downarrow \\ & + \text{Business investment (I)} \\ & + \text{Government spending (G)} & \uparrow \\ & + \text{Net exports (NX)} \end{aligned}$$

Keynes argued that in a recession, when consumption falls, governments should make up for this lost demand by purchasing goods themselves. Keynes argued that this would stabilize GDP. The crucial question here is whether a dollar spent by the government increases GDP by more than \$1.

I

$>\$1$

 GDP = Private consumption (C) 
+ Business investment (I)
+ Government spending (G)  \$1
+ Net exports (NX)

Government spending multiplier >1

This is called the government spending multiplier. In the Keynesian view, the government spending multiplier is larger than one, at least in recessions. This is because the government makes up for lost demand, which creates jobs and eventually increases consumption.

I

$<\$1$

 GDP = Private consumption (C) 
+ Business investment (I)
+ Government spending (G)  \$1
+ Net exports (NX)

Government spending multiplier <1

Opponents of the Keynesian view argue that the government spending multiplier is less than one. They argue that the government cannot simply substitute for private

consumption. They suggest that the government replaces private demand with demand for different types of goods. Think about when the government increases military spending in response to a shock originating in the housing sector. Workers from the construction sector would need to find employment in the defense sector. This is clearly problematic as construction workers may not have the skills needed in the defense industry. Retooling workers in reallocating them across sectors can be costly and inefficient. Hence, the argument goes that the government spending multiplier is less than one. While most studies indicate that during recessions the government spending multiplier is larger than one, the evidence is mixed. As you might expect, the answer usually depends on how the money is spent.



In a recession, the government should spend immediately but often projects take time.

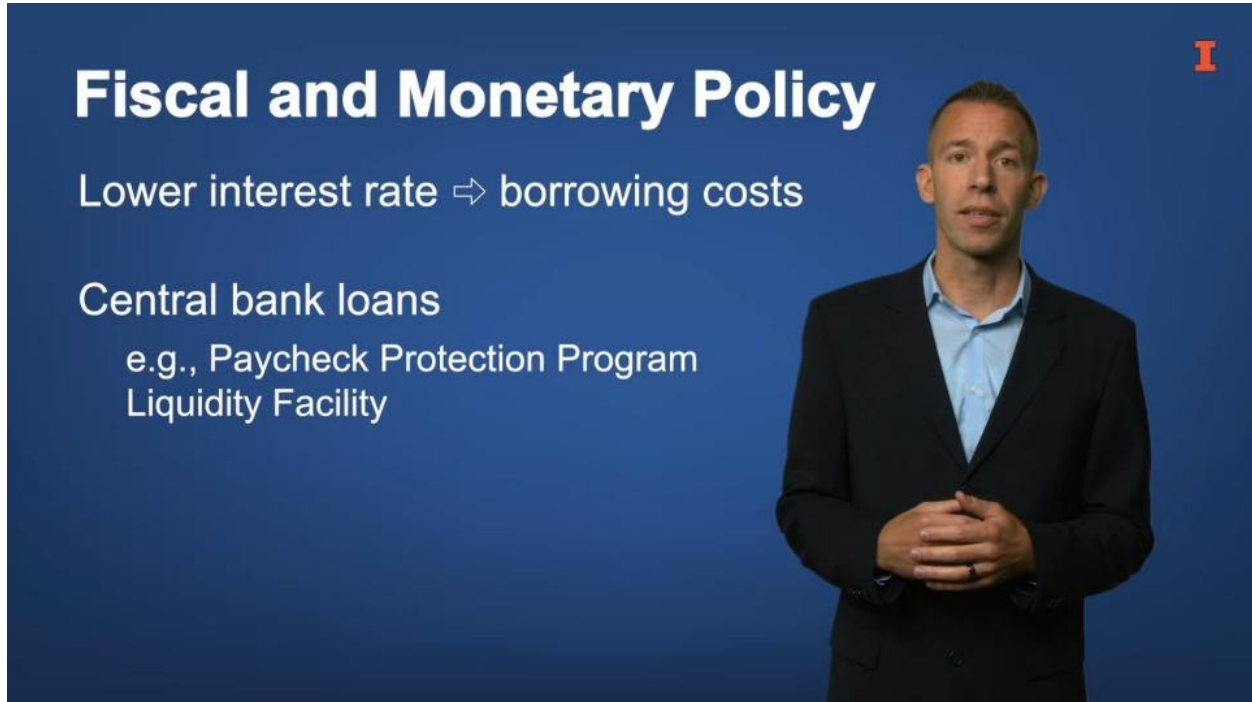


Think about large infrastructure projects. They need careful planning, buy in from the stakeholders, and to go through the government purchasing process. All of this means that the implementation of programs is likely to be delayed.



It is often hard to measure the effects of this spending. How do you measure the benefit of a repaired bridge? In sum, under some conditions, increasing government spending can be an effective way to support the economy in a recession. One final thought before

we wrap up. Monetary policy can play an important role in supporting these fiscal responses. How? Well, no matter which of these three types of government intervention is implemented, they all cost money. Remember that the central bank controls the interest rate. By lowering the interest rate, they can support a government stimulus program in two ways.



Fiscal and Monetary Policy

Lower interest rate ⇒ borrowing costs

Central bank loans
e.g., Paycheck Protection Program
Liquidity Facility

First, it directly lowers the borrowing costs of the government. Second, it becomes cheaper for businesses to invest and for households to take out loans for consumption. This can amplify the effect of the government stimulus. The central bank can also support a stimulus program by providing funding for financial institutions. A leading example is the Paycheck Protection Program Liquidity Facility in 2020. Under this program, lenders receiving funding from the central bank to support grants made to small businesses under the government's Paycheck Protection Program. Without this central bank's support, lenders facing their own constraints might have been reluctant to help with this government program.

Summary

How to prevent a terrible recession?

Government transfers and spending

Targeted interventions work best

Monetary policy can help



Let's wrap up, this was a session on the fiscal response to financial crisis. The central issue is how the government can support the economy to avoid the worst-case scenario. We covered three ideas. First, the government can prop up demand by either transferring money to households or increase government and spending. Second, targeted interventions, such as subsidizing wage bills or increasing unemployment benefits, are the most effective. Third, supportive monetary policy can help to facilitate government stimulus programs.